

Context free grammar (CFG)

CFG: 4-tuples :- $G = \{V, T, P, S\}$

where V is set of variables or non terminals upper case letter. (substitution)

T is set of terminals or symbols lower case letter. (generate string)

P is set of production Rule.

$A \rightarrow \alpha$ where $A \in V$, $\alpha \in (V \cup T)^*$.

S is start symbol (always non terminal)

Ex:- consider the grammar, $E \rightarrow E+E \mid E * E \mid (E) \mid id$
to obtain $id + id * id$

$E \rightarrow E+E$ ($E \rightarrow E+E$)

$E \rightarrow id+E$ ($E \rightarrow id$)

$E \rightarrow id+E * E$ ($E \rightarrow E * E$)

$E \rightarrow id + id * id$ ($E \rightarrow id$)

Derivation: left most derivation, RMD

Derivation: application of sequence of various production rules to obtain a particular string

left most Derivation: each step left most non terminal

has to be expanded.

Right most Derivation: Right most non terminal

has to be expanded.

Derivation tree / parse tree :- Diagrammatic rep of

Derivation process.

Rules

- 1) Root node $\rightarrow$ start symbol
- 2) Internal node $\rightarrow$ non terminal nodes
- 3) leaf nodes $\rightarrow$ terminal nodes.

Ex:- $E \rightarrow E + E \mid E * E \mid (E) \mid id$

LMD $\hookrightarrow id + id * id \rightarrow$ obtain
RMD $\leftarrow$

$E \rightarrow E + E$

$E \rightarrow id + E$

$E \rightarrow id + id * E$

$E \rightarrow id + id * E$

$E \rightarrow id + id * id$

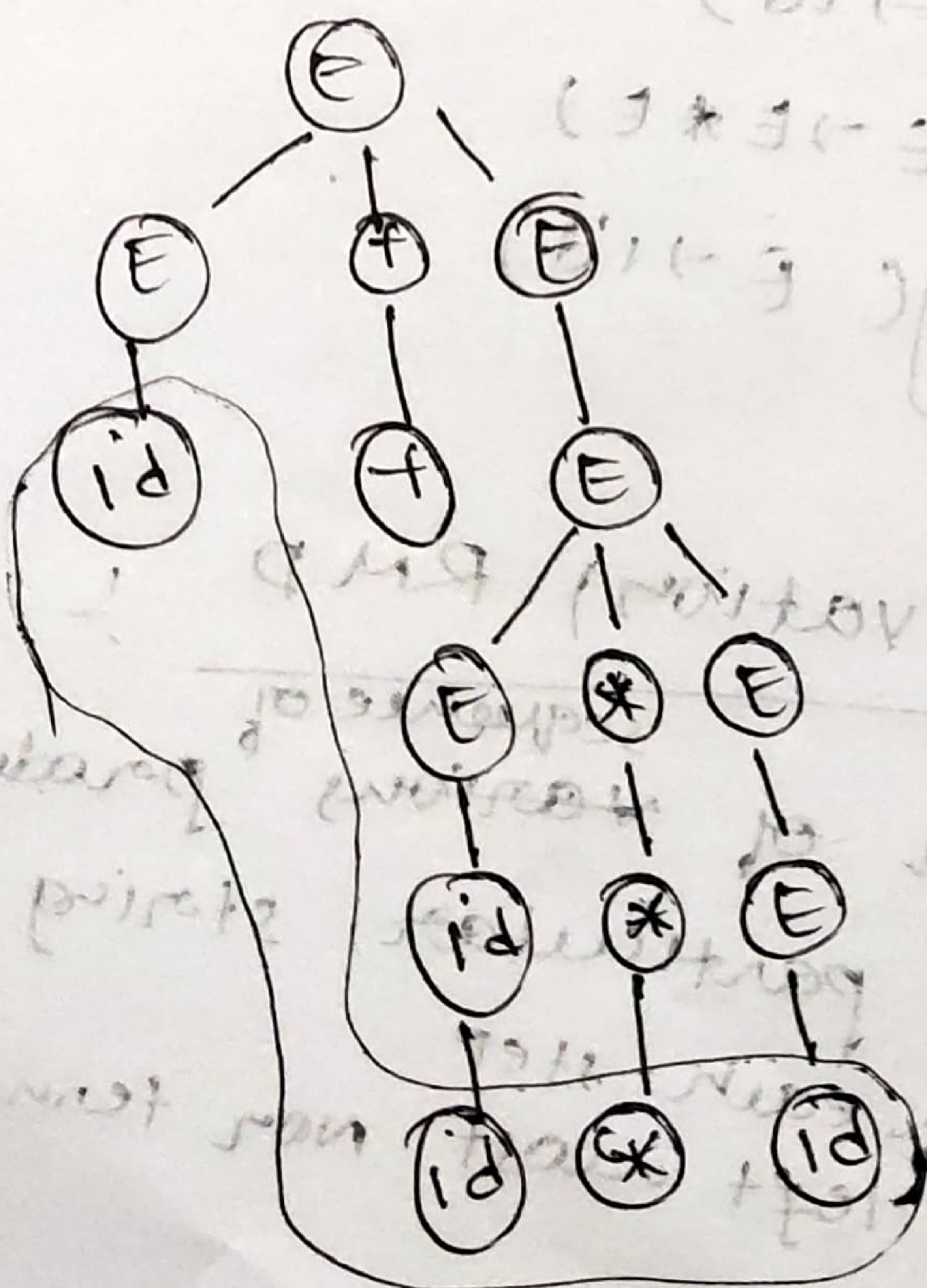
$E \rightarrow E + E$

$E \rightarrow E + E * E$

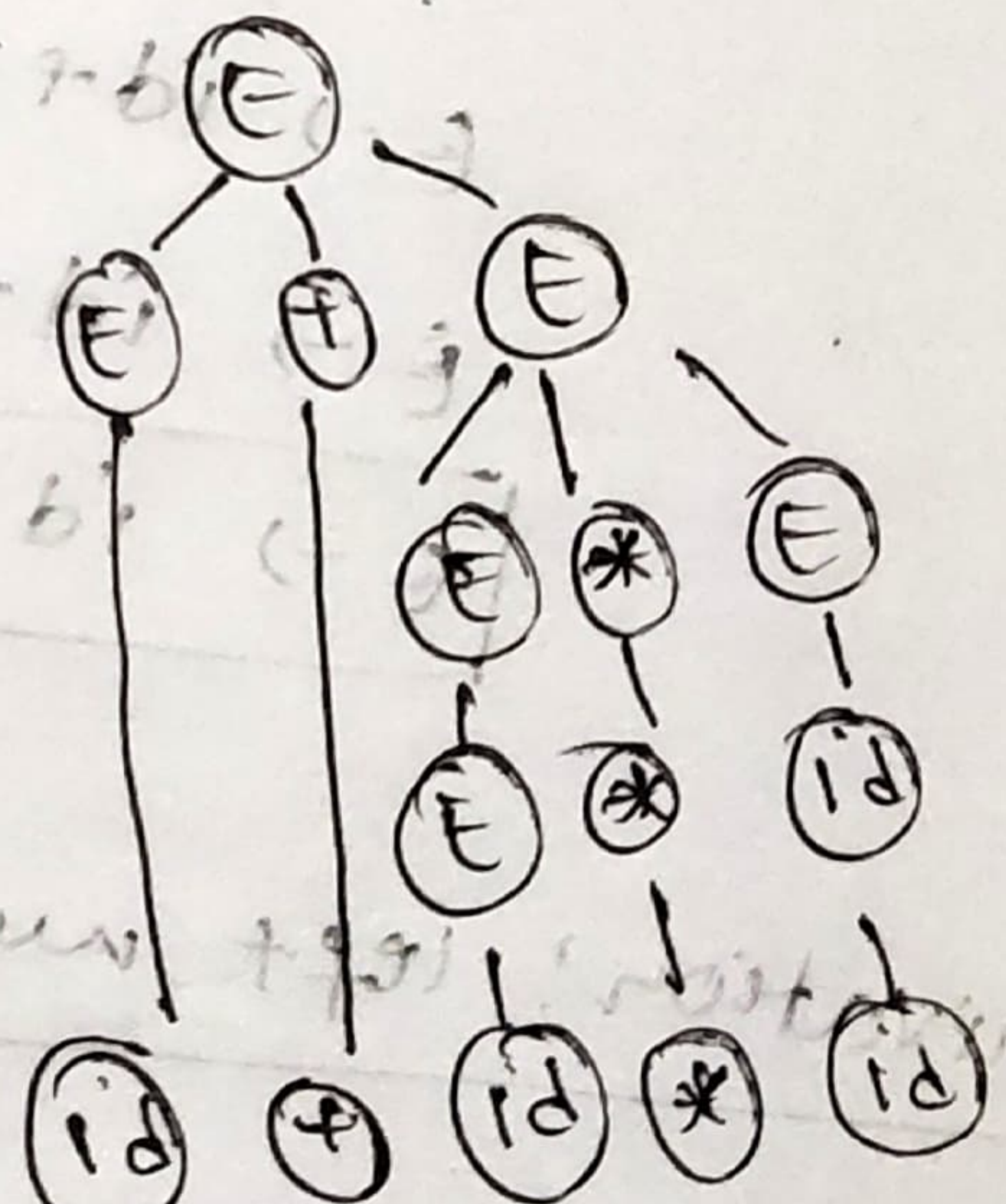
$E \rightarrow E + E * id$

$E \rightarrow E + id * id$

$E \rightarrow id + id * id$



parse tree left MD



construction of CFG for a language

Any no of a's over set $E = \{a\}$.

$$L = \{ \epsilon, a, aa, \dots \}$$

$$R = a^*$$

production rules

$$\left. \begin{array}{l} s \rightarrow as \\ s \rightarrow \epsilon \end{array} \right\} \boxed{s \rightarrow as | \epsilon}$$

$$G = \{ \{s\}, \{a\}, P, S \}$$

$$\begin{array}{l} aaa \\ s \rightarrow as \\ s \rightarrow aas \\ s \rightarrow aaas \\ \underline{s \rightarrow aaat} \end{array}$$

Any no of a's & b's

$$E = \{a, b\}$$

$$L = \{ \epsilon, a, b, ab, aab \}$$

$$P = (a+b)^*$$

$$aab$$

$$\begin{array}{l} s \rightarrow as \\ s \rightarrow aas \\ s \rightarrow aabs \\ s \rightarrow aab\epsilon \\ s \rightarrow aab \end{array}$$

$$\boxed{P: s \rightarrow as | bs | \epsilon}$$

$$\boxed{G = \{ \{s\}, \{a, b\}, P, S \}}$$

containing strings of atleast 2 a's.

$$L = \{ aa, aaa, aas, \dots \}$$

$$R = aa (a+b)^* a (a+b)^* a$$

$$P: s \rightarrow AaAa$$

$$A \rightarrow aA | bA | \epsilon$$

$$\boxed{G = \{ \{s, A\}, \{a, b\}, P, S \}}$$

4) one occurrence of 0001: in strings

$$(0+1)^* 000 (0+1)^*$$

P:

$$S \rightarrow A000A$$

$$A \rightarrow 0A1A1 \epsilon$$

$$G = (\{S, A\}, \{0, 1\}, P, S)$$

5) set of strings with equal no of a's & b's.

$$L = \{ab, ba, a^2b^2, b^2a^2, \dots\}$$

$$S \rightarrow aSbS \mid bSaS \mid \epsilon$$

$$\cancel{S \rightarrow aS}$$

$$S \rightarrow aSbS$$

$$S \rightarrow a \epsilon bS$$

$$S \rightarrow abS$$

$$S \rightarrow aSbS$$

$$S \rightarrow aSb \epsilon a \epsilon$$

$$S \rightarrow aSb a$$

6) construction of CFN for $(0+1)^*$

$$L = \{\epsilon, 011, 01, 001, 0111, \dots\}$$

$$P: S \rightarrow 0S1S1 \epsilon$$

$$G = (\{S\}, \{0, 1\}, P, S)$$

7)

pairing none over (a^2b^2)

$$\epsilon = \{a^2b^2\}$$

$$L = \{\epsilon, aa, bb, ab, aaba, baab, bbaa, \dots\}$$

P: $s \rightarrow asa | ssb | e | a | b$ (even)

$s \rightarrow asa | bsb | e$ (even)

$s \rightarrow asa | bsb | a | b$ (odd)

assa

$s \rightarrow asa$

$s \rightarrow assb a$

$s \rightarrow abesa$

$s \rightarrow assb a$

8) CFG for $L = \{ w w^R | w \in \{a, b\}^* \}$

~~odd~~ $\Rightarrow s \rightarrow asa | bsb | e$

exactly one 'a' $\{a, b\}^*$

$s^* a b^*$

P: $s \rightarrow BAB$

$B \rightarrow b B | e$

9) at least one 'a' $\{a, b\}^*$

$(a+b)^* a (a+b)^*$

P: $s \rightarrow A A A$

$A \rightarrow a A | b A | e$

11) $L = \{ 0^n 1^n | n \geq 0 \}$

$L = \{ \epsilon, 01, 0011, 000111, \dots \}$

~~$s \rightarrow 0s | 1s | e$~~

$s \rightarrow 0s | 1e$

$$12) L = \{ a^n b^n \mid n \geq 1 \}$$

$$L = \{ a^n b^n \mid n \geq 0 \}$$

$$P: s \rightarrow a s b \mid a b$$

$$13) L = \{ 0^n \mid n \geq 0 \}$$

$$L = \{ \epsilon, 0, 00, \dots \}$$

$$P: s \rightarrow \epsilon \mid 0 s$$

$$14) L = \{ a^m b^n \mid m \geq 0, n \geq 0 \}$$

$$L = \{ \epsilon, a, b, a b, a a b, a b b, \dots \}$$

$$s \rightarrow \epsilon \mid a s \mid b s \mid \text{Wrong}$$

$s \rightarrow a A$
 $A \rightarrow a A \mid \epsilon$
 $B \rightarrow b B \mid \epsilon$

$$15) L = \{ a^m b^n \mid m \geq 0, n \geq 1 \}$$

$$L = \{ b, a b b, a a b b b, \dots \}$$

$a b b$
 $a s$
 $a b s$
 $a b b$

$$s \rightarrow a s \mid b s \mid b$$

Wrong

$s \rightarrow A B$
 $A \rightarrow a A \mid \epsilon$
 $B \rightarrow b B \mid b$

$s \rightarrow a s$
 $s \rightarrow a a s$
 $s \rightarrow a a b s$
 $s \rightarrow a a b b s$
 $s \rightarrow a a b b b$

$$16) L = \{a^n b^{2n} \mid n \geq 0\}$$

$$L = \{\epsilon, abb, aabbbb, \dots\}$$

$$s \rightarrow asbb \mid \epsilon$$

$$17) L = \{a^n b^m \mid n \geq 0, m \geq 0\}$$

$$L = \{\epsilon, aab, aaaaab, \dots\}$$

$$\cancel{s \rightarrow \epsilon \mid as \mid bs}$$

$$s \rightarrow aasb \mid \epsilon$$

$$aaaaab$$

$$s \rightarrow a$$

$$s \rightarrow aa$$

$$s \rightarrow aaaa$$

$$s \rightarrow aaaaab$$

$$s \rightarrow aaaaasb$$

$$0^* 1 (0+1)^*$$

$$A^* 1 B^*$$

$$A \rightarrow 0A \mid \epsilon$$

$$B \rightarrow 0A \mid 01 \mid \epsilon$$

$$19) L = \{w \mid n_a(w) = n_b(w)\}$$

$$L = \{\epsilon, ab, aabb, aabbbb, bbaa, bbaabb, \dots\}$$

$$s \rightarrow asbs \mid bsas \mid \epsilon$$

$$20) L = \{a^n b^{n+2} \mid n \geq 0\}$$

$$s \rightarrow bb \mid asb$$

$$L = \{bb, abbb, aabbbb, \dots\}$$

Simplification of CFG

$$\begin{array}{c} A \rightarrow B \\ \text{NT} \quad \text{NT} \end{array} \quad \downarrow$$

$$A \rightarrow x_1 x_2 x_3 \dots x_n$$

Ex:

$$S \rightarrow 0A1B1C$$

$$A \rightarrow 0S100$$

$$B \rightarrow 1A$$

$$C \rightarrow 01$$

unit production

$$S \rightarrow C$$

$$B \rightarrow A$$

Eliminate $S \rightarrow C, C \rightarrow 01$

$$\underline{S \rightarrow 01}$$

Eliminate $B \rightarrow A, A \rightarrow 0S100$

$$B \rightarrow 0S100$$

final grammar

$$S \rightarrow 0A1B101$$

$$A \rightarrow 0S100$$

$$B \rightarrow 0S100$$

$$C \rightarrow 01$$

Ex2:

$$S \rightarrow AB$$

$$A \rightarrow a$$

$$B \rightarrow c/b$$

$$C \rightarrow D$$

$$D \rightarrow E/bC$$

$$E \rightarrow dAB$$

unit prod:

$$B \rightarrow C$$

$$C \rightarrow D$$

$$D \rightarrow E$$

$B \rightarrow D$
 $D \rightarrow E, E \rightarrow d|ab$
 $D \rightarrow d|ab|bc$
 $C \rightarrow D, D \rightarrow \cancel{d|ab|bc} E$
 $C \rightarrow E$
 $C \rightarrow d|ab$
 $B \rightarrow C, C \rightarrow D \Rightarrow B \rightarrow D$
 $B \rightarrow d|ab|bc|$

final prod:

$S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow d|ab|bc|b$
 $C \rightarrow d|ab|bc$
 $D \rightarrow d|ab|bc$
 $E \rightarrow d|ab$

removal of unit productions

If any production is in the form $A \rightarrow B$
 then it is called unit production.
 To eliminate unit productions, we follow
 rule

If a production is of the form $A \rightarrow B$
 & $B \rightarrow x_1 x_2 \dots x_n$ then write the following
 productions as $A \rightarrow x_1 x_2 \dots x_n$

Ex: $S \rightarrow Aa|B$
 $B \rightarrow A|bb$
 $A \rightarrow a|bc|B$

★

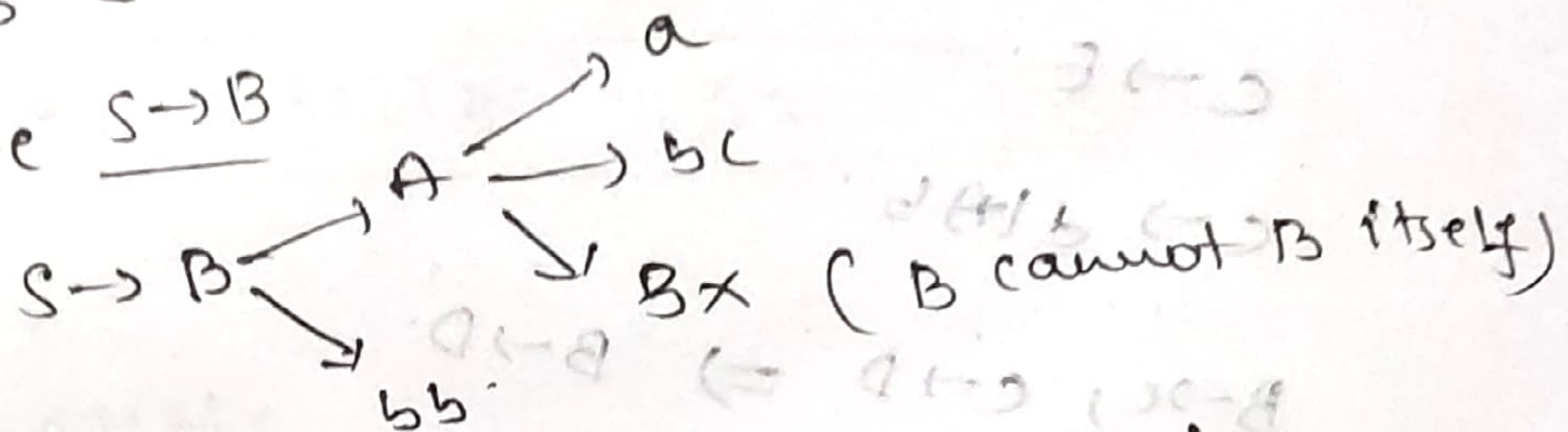
$S \rightarrow B$

$B \rightarrow A$

$A \rightarrow B$

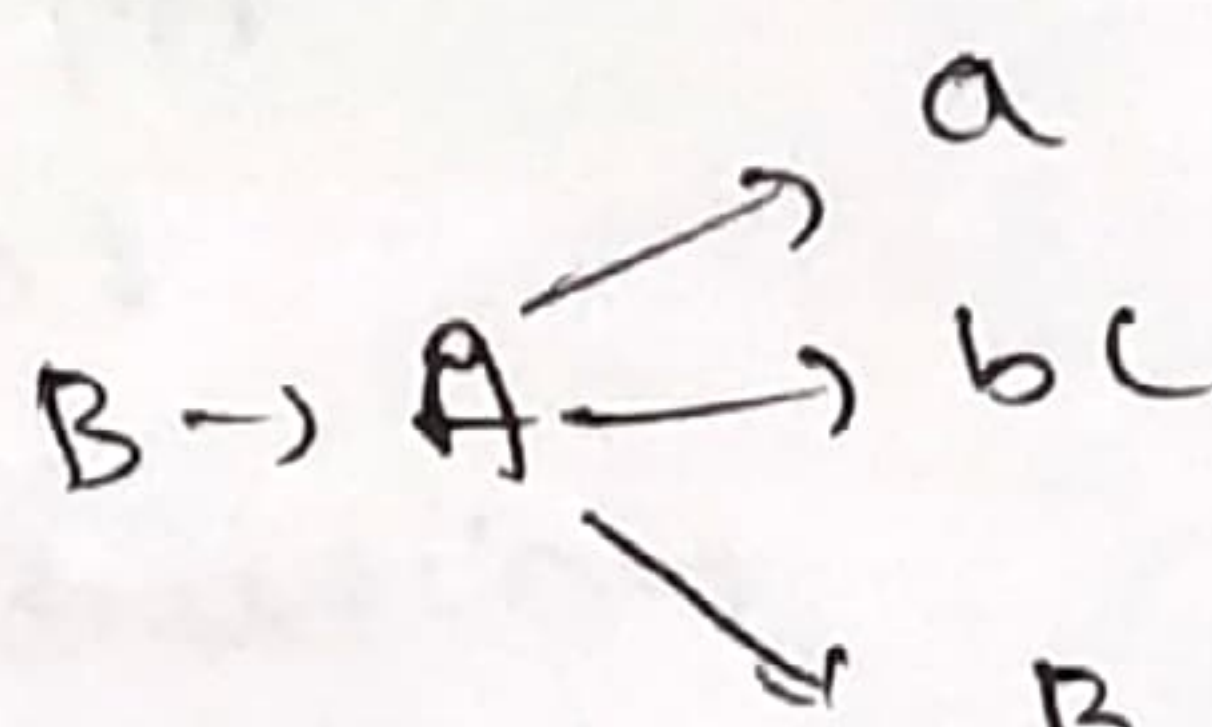
Imp because in this all productions have unit productions. Every nt prod every other nt

Eliminate $S \rightarrow B$



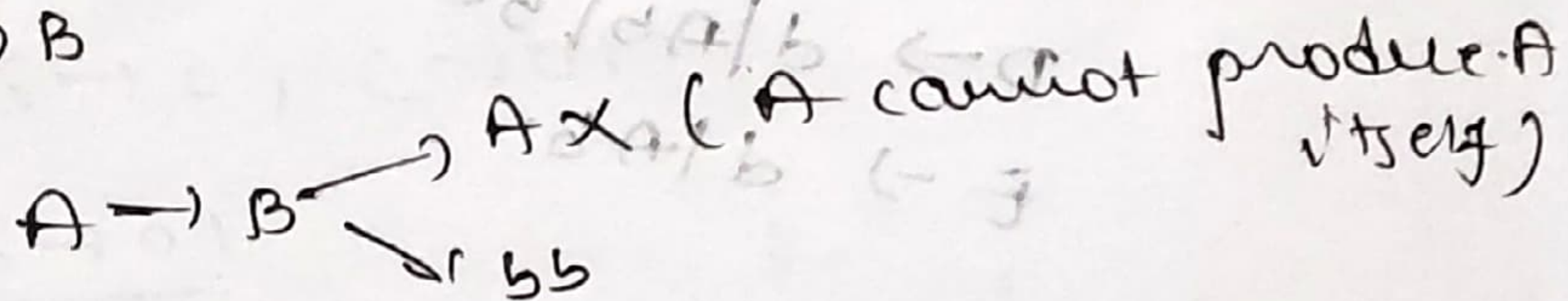
$S \rightarrow Aa|a|bc|bb$

Eliminate $B \rightarrow A$



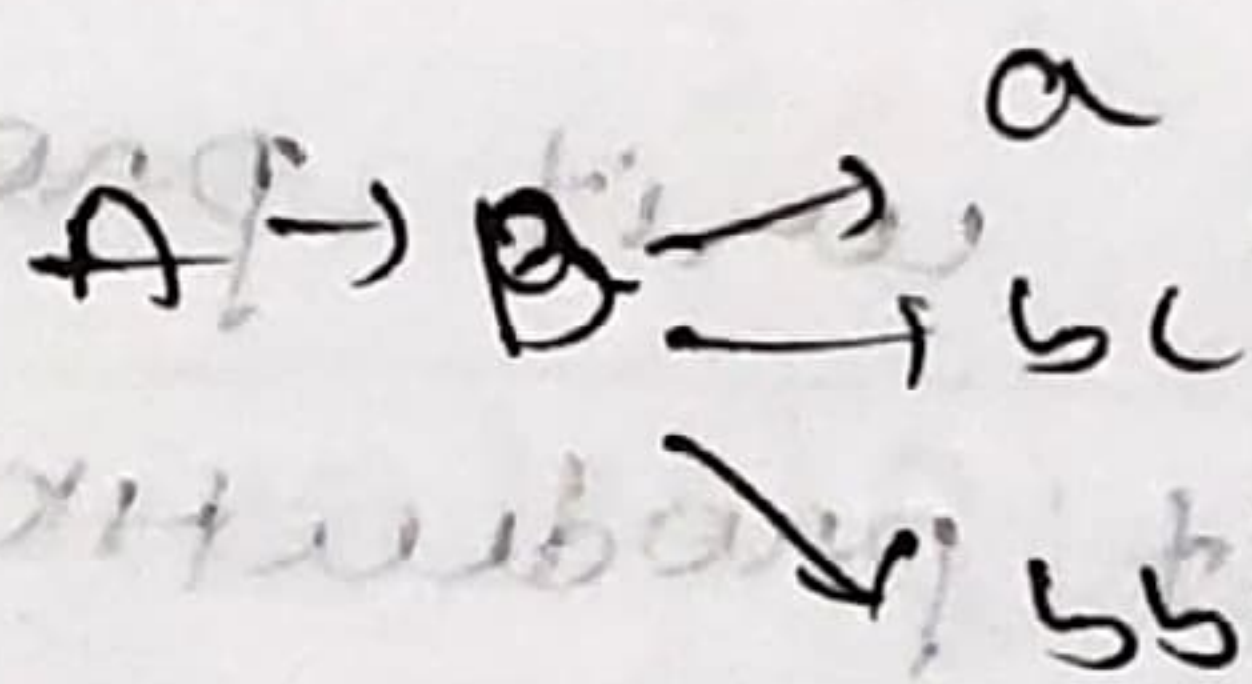
$B \rightarrow a|bc|bb$

Eliminate $A \rightarrow B$



$A \rightarrow a|bc|bb$

Other way



$A \rightarrow a|bc|bb$

final production:

$S \rightarrow Aa|a|bc|bb$

$B \rightarrow a|bc|bb$

$A \rightarrow a|bc|bb$

Removal of Null production

$A \rightarrow \epsilon$ null production.

Ex: $S \rightarrow x y x$ null prod

$x \rightarrow 0 x 1 \epsilon$

$x \rightarrow \epsilon$

$y \rightarrow 1 y 1 \epsilon$

$y \rightarrow \epsilon$

Eliminate $x \rightarrow \epsilon$ replace each occurrence

of x with ϵ .

$S \rightarrow x y x \mid x y \mid y x \mid y$

$x \rightarrow 0 x 1 \mid 0$

$y \rightarrow 1 y 1 \mid \epsilon$

Eliminate $y \rightarrow \epsilon$ replace each occurrence of y with ϵ .

$S \rightarrow x y x \mid x y \mid y x \mid y \mid x x \mid x \mid \epsilon$

$x \rightarrow 0 x 1 \mid 0$

$y \rightarrow 1 y 1 \mid \epsilon$

Ex:

$A \rightarrow 0 B 1 \mid 1 B 1$

$B \rightarrow 0 B 1 \mid 1 B 1 \mid \epsilon$

Null prod

$B \rightarrow \epsilon$

$A \rightarrow 0 B 1 \mid 1 B 1 \mid 0 1 \mid 1 1$

$B \rightarrow 0 B 1 \mid 1 B 1 \mid 0 1 \mid 1 1$

Removal of useless symbols $\rightarrow$ non terminals.

- 1) Non terminal must not be derived from starting symbol.
- 2) If a nonterminal can't produce a string of terminals.

ex: $S \rightarrow xy10$ $y \rightarrow \text{unkey.}$

~~$s \rightarrow x + 0$~~ remove $s \rightarrow xy$

$z \rightarrow 0$ ✓
 $x \rightarrow 1$ ✓

የገሥት

$$S \rightarrow A \cup B \mid \cup A$$

5- 11B 1 11P 1 11P 1

$A \rightarrow 0$

$B \rightarrow BB$ $\times$ (not produce string of terminals)

5-11A.

5-111

A-10

final

(-) $A \cup B$ $\begin{cases} \text{we} \\ \text{we} \end{cases}$

→ 11B

Reduction / minimization of UFG

$S \rightarrow A/OCV$

$A \rightarrow B^{101/10}$

$$c \rightarrow e/cD$$

S1) elimination of useless symbols

$S \rightarrow A \mid 0 \mid 1$

$S \rightarrow 0 \mid 1$

$S \rightarrow 0 \mid 1$

$A \rightarrow B \mid X$

$A \rightarrow 0 \mid 1$

$A \rightarrow 1 \mid 0$

$C \rightarrow \epsilon$

$C \rightarrow C \mid D$

After eliminating useless sym,

$S \rightarrow A \mid 0 \mid 1$

$A \rightarrow 0 \mid 1 \mid 10$

$C \rightarrow \epsilon$

S2: Elimination of ϵ production

$C \rightarrow \epsilon$

$S \rightarrow A \mid 0 \mid 1$

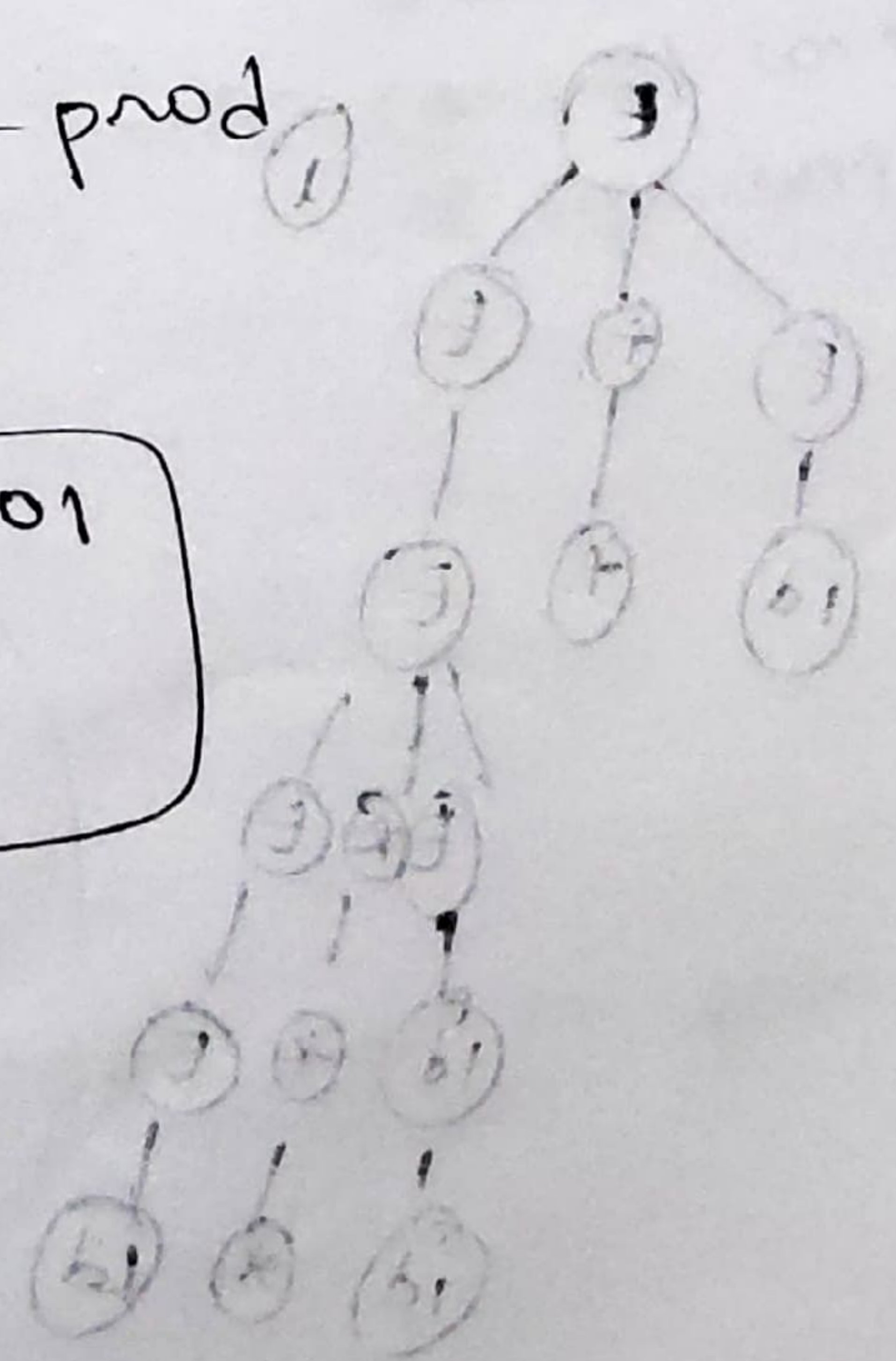
$A \rightarrow 0 \mid 1 \mid 10$

S3: Elimination of unit production

$S \rightarrow A \rightarrow \text{unit prod}$

$A \rightarrow 0 \mid 1 \mid 10$

$S \rightarrow 0 \mid 1 \mid 10 \mid 00 \mid 01 \mid 10$
 $A \rightarrow 0 \mid 1 \mid 10$



Ambiguous Grammar:

A grammar said to be ambiguous if there exists more than 1 parse tree for the given I/P string.

- more than 1 left most parse tree.
- ... Right most parse tree.
- combination of both (one left, one right)

Ex:- $E \rightarrow E + E \mid E * E \mid id$

① left most:-

$E \rightarrow E + E$

$E \rightarrow id + E$

$E \rightarrow id + E * E$

$E \rightarrow id + id * E$

$E \rightarrow id + id * id$

②

right most:-

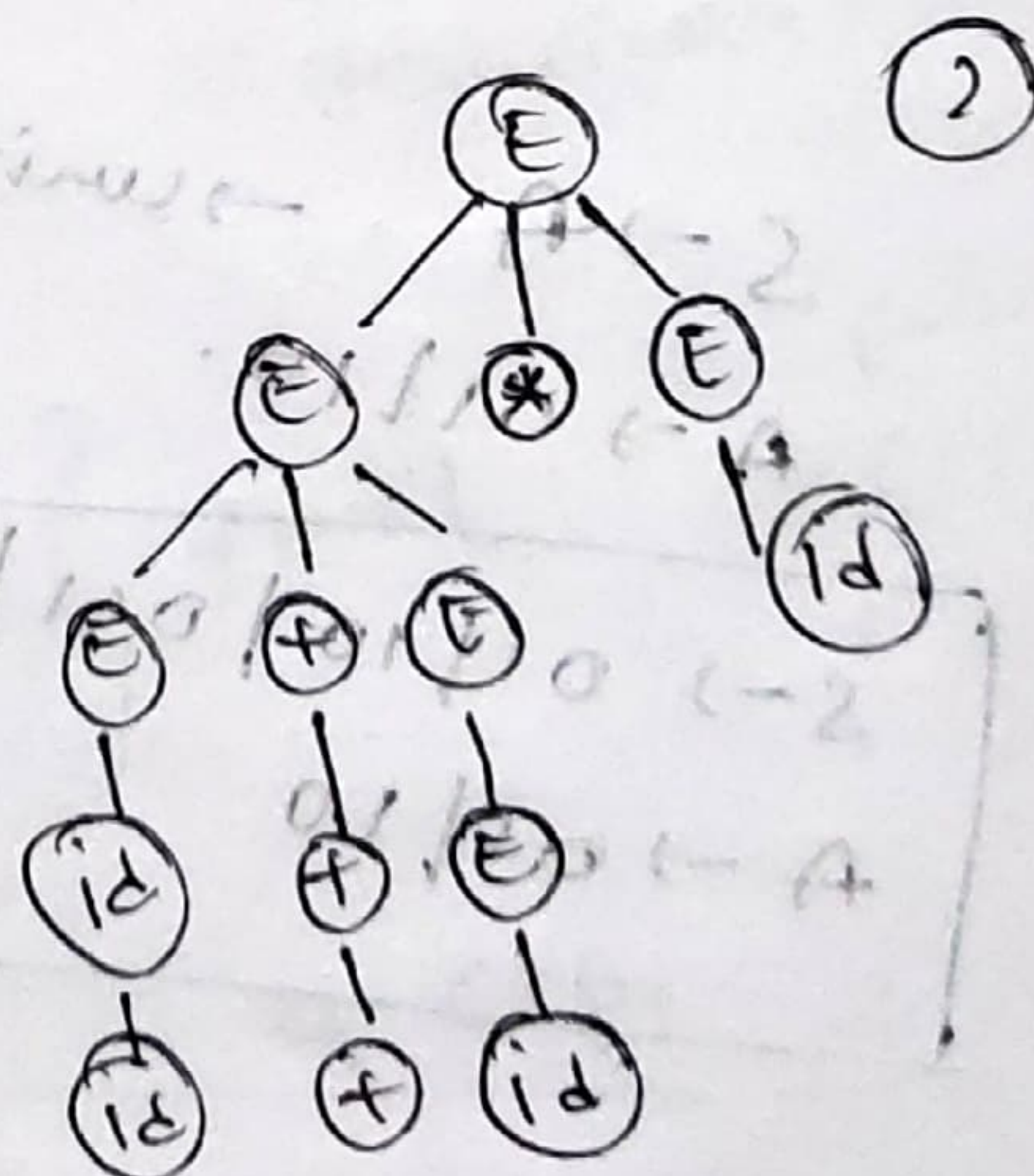
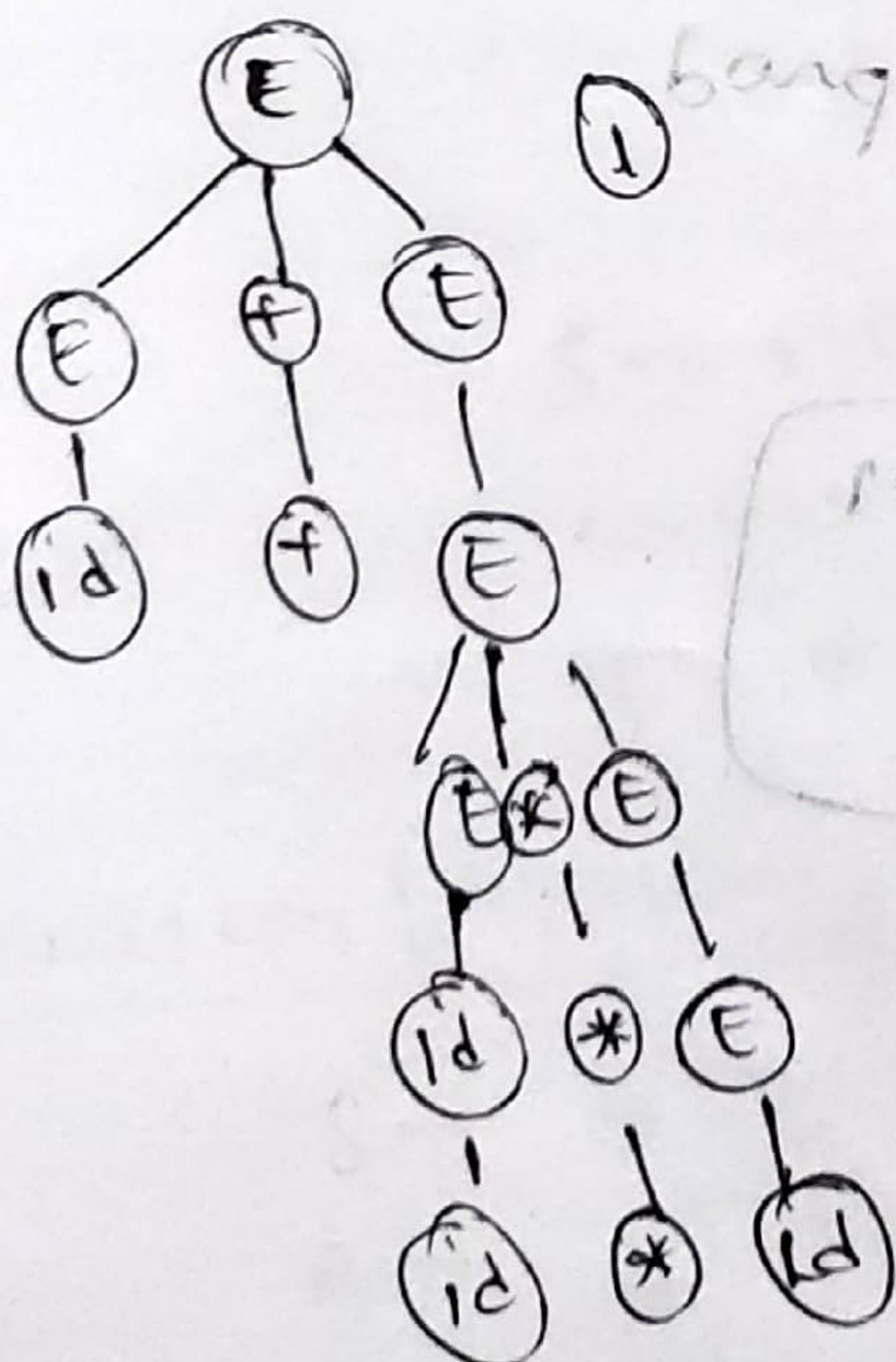
$E \rightarrow E * E$

$E \rightarrow E + E * E$

$E \rightarrow id + E * E$

$E \rightarrow id + id * E$

$E \rightarrow id + id * id$



Since we have 2 left most parse trees for same gram it is ambiguous.

Elimination of left Recursion Removal of ambiguity in grammar

1) left recursion
 $A \rightarrow A\alpha \mid \beta$

2) right recursion
 $A \rightarrow \alpha A \mid \beta$

$A()$
 $\{$
 $A();$
 $\alpha \dots$
 $\}$

we only execute $A()$ not α this is the problem.

$A()$
 $\{$
 $\alpha;$
 $A();$
 $\}$

Here α will be executed then $A()$

* Top down parsing techniques cannot handle left recursion.

So no problem in Right Recursion it occurs mainly only in left Recursion.

So we use

$A \rightarrow BA'$
 $A' \rightarrow \alpha A' \mid \epsilon$ } to eli left recursion.

or above ex:

A
 $\mid$
 B

 B

A
 $\mid$
 $A \alpha$
 $\mid$
 B

 $B\alpha$

A
 $\mid$
 A
 $\mid$
 $A \alpha$
 $\mid$
 B

 $B\alpha\alpha$

But grammar cannot contain * symbol

So we use,

$A \rightarrow BA'$
 $A' \rightarrow \epsilon \mid \alpha A'$

Ex1: Eliminate left recursion for the following grammar

$$Q) E \rightarrow E + T \mid T$$

$$T \rightarrow T * F \mid F$$

$$F \rightarrow (E) \mid id$$

$$\underline{E} \rightarrow \underline{E} + \underline{T} \mid \underline{T} \rightarrow \text{left recursion.}$$

$$A : A \alpha B$$

$$\begin{cases} E \rightarrow TE' \\ E' \rightarrow +TE' \mid \epsilon \end{cases}$$

$$T \rightarrow T * F \mid F$$

$$A \quad A \quad B \quad B$$

$$\begin{cases} T \rightarrow FT' \\ T' \rightarrow *FT' \mid \epsilon \end{cases}$$

$$F \rightarrow (E) \mid id \rightarrow \text{not left R.}$$

final step

~~$$E \rightarrow E + T$$~~

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' \mid \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' \mid \epsilon$$

$$F \rightarrow (E) \mid id$$

$$S \rightarrow SOSIS \mid OI$$

$$A \quad A \quad \alpha \quad B$$

$$\begin{cases} S \rightarrow OIS' \\ S' \rightarrow OSISS' \mid \epsilon \end{cases}$$

Q3) $S \rightarrow (L) | \neg L \rightarrow \text{not } LP$
 $L \rightarrow L1S | S$

$S \rightarrow (L) | \neg L \rightarrow \text{not } LP$

$L \rightarrow L1S | S$

$\bar{A} \quad \bar{A} \alpha \quad B$

$L \rightarrow SK'$
 $L' \rightarrow ,SK' | \epsilon$

concept:

$A \rightarrow A\alpha_1 | A\alpha_2 | \dots | B_1 | B_2 | \dots$

$A \rightarrow \beta_1 A' | \beta_2 A' | \dots$

$A' \rightarrow \alpha_1 A' | \alpha_2 A' | \dots | \epsilon$

4Q) $\text{expr} \rightarrow \text{expr} \underbrace{+ \text{expr}}_{\alpha_1} | \text{expr} \underbrace{* \text{expr}}_{\alpha_2} | \underbrace{\text{id}}_B$

$\text{expr} \rightarrow \text{id expr}'$

$\text{expr}' \rightarrow + \text{expr expr}' | * \text{expr expr}' | \epsilon$

5Q) $S \rightarrow S \underline{a} | S \underline{b} | \underline{X S} | \underline{a}$
 $\bar{A} \quad \bar{A} \alpha_1 \quad \bar{A} \alpha_2 \quad B_1 \quad B_2$

$S \rightarrow X S S' | a S'$

$S' \rightarrow X S' | S b S' | \epsilon | a S'$

6Q)

$S \rightarrow Aa | b$

$A \rightarrow AC | Sd | \epsilon$

~~$A \rightarrow AC | Sd | \epsilon$~~

~~$\bar{A} \quad A$~~

$$A \rightarrow AC \mid sd \mid f$$

direct
LP

Indirect

LP (substitution of prod
in to other)

Replace s with Aa

$$A \rightarrow AC \mid Aad \mid f \mid bd$$

left rec

In case of Indirect LP

1st order of nonterminals S, A

So substitute S production in A

$$A \rightarrow AC \mid Aad \mid Bd \mid f$$

$$A \rightarrow bdA' \mid fA'$$

$$A' \rightarrow CA' \mid adA' \mid \epsilon$$

$$S \rightarrow Aab$$

$$A \rightarrow bdA' \mid fA'$$

$$A' \rightarrow CA' \mid adA' \mid \epsilon$$

Elimination of left factoring!

$$A \rightarrow \alpha B_1 \mid \alpha B_2 \mid \dots \mid r_1 \mid r_2 \dots$$

problem! - the top down parsers cannot
handle the grammar which have left
factoring
we solve this prob using production,

$$A \rightarrow aA \mid \epsilon$$

$$A \rightarrow B \mid B^2 \mid B^3 \mid \dots$$

$$S \rightarrow ietss \mid ieties \mid a$$

$$\begin{aligned} S &\rightarrow ietss \mid a \\ S' &\rightarrow e \mid es \\ E &\rightarrow b \end{aligned}$$

→ final

Q2) $A \rightarrow aAB \mid aA \mid a$

$$B \rightarrow bB \mid b$$

$$A \rightarrow \underbrace{aAB}_{\alpha \beta_1} \mid \underbrace{aA}_{\alpha \beta_1} \mid a$$

$$B \rightarrow \underbrace{bB}_{\alpha \beta_1} \mid \underbrace{b}_{\alpha}$$

$$\begin{aligned} A &\rightarrow aA \mid \epsilon \\ A &\rightarrow AB \mid a \mid \epsilon \end{aligned}$$

$$\begin{aligned} A &\rightarrow bB' \\ A &\rightarrow B \mid \epsilon \end{aligned}$$

30) $X \rightarrow x+x \mid x*x \mid D$

$$D \rightarrow 1 \mid 2 \mid 3$$

$$X \rightarrow \underbrace{x+x}_{\alpha \beta_1} \mid \underbrace{x*x}_{\alpha \beta_2} \mid \underbrace{D}_{\alpha \beta_3}$$

$$\begin{aligned} A &\rightarrow xA' \mid D \\ A' &\rightarrow +x \mid *x \\ D &\rightarrow 1 \mid 2 \mid 3 \end{aligned}$$

→ final

Q4)

$$E \rightarrow TE | T$$

$$T \rightarrow \text{int} | \text{int} * T | (E)$$

$$E \rightarrow \underbrace{TE}_{\alpha \beta_1} | \underbrace{T}_{\alpha} \quad T \rightarrow \underbrace{\text{int}}_{\alpha} | \underbrace{\text{int} * T}_{\alpha \beta_2} | \underbrace{(E)}_{\alpha}$$

$$\begin{aligned} E &\rightarrow TE' \\ E' &\rightarrow TE | \epsilon \end{aligned}$$

$$\begin{aligned} T &\rightarrow \text{int} T' | (E) \\ T' &\rightarrow \epsilon | * T \end{aligned}$$

Chomsky Normal form (CNF)

A CFG is said to be in CNF if productions are of the form $A \rightarrow BC$ where A, B, C are $\in T$ and a is terminal.

Ex:-

$$S \rightarrow SA | aB$$

$$A \rightarrow SA | a | a$$

$$B \rightarrow aBB | bB | b$$

$$A \rightarrow a$$

$$B \rightarrow b$$

} already in CNF

find equivalent CNF

S1: first minimize the given CFG

S2: replace all terminals on RHS with non-terminals.

$$S \rightarrow bA$$

$$S \rightarrow (bA$$

$$(b \rightarrow b$$

$$A \rightarrow aS$$

$$A \rightarrow (aS$$

$$(a \rightarrow a$$

$$S \rightarrow aB$$

$$S \rightarrow (aB$$

$$(a \rightarrow a$$

$$B \rightarrow aBB$$

$$B \rightarrow (aBB$$

$$(a \rightarrow a$$

$$A \rightarrow bAA$$

$$S \rightarrow (bAA$$

$$(b \rightarrow b$$

$$B \rightarrow bS$$

$$B \rightarrow (bS$$

$$(b \rightarrow b$$

$A \rightarrow CbAA$
 $A \rightarrow CbD_1$
 $D \rightarrow AA$

$B \rightarrow CaBB$
 $B \rightarrow CaD_2$
 $D_2 \rightarrow BB$

they were not
 in CNF even
 after 1st itera

$S \rightarrow CbA$
 $Cb \rightarrow b$
 $S \rightarrow CaB$
 $Ca \rightarrow a$
 $A \rightarrow CbD_1$
 $D \rightarrow AA$
 $A \rightarrow Cas$
 $B \rightarrow CaD_2$
 $D_2 \rightarrow BB$
 $B \rightarrow Cbs$

→ CNF
 form. equivalent to given
 question

20) $S \rightarrow a|aA|B$
 $A \rightarrow aBB|E$
 $B \rightarrow Aa|b$

eliminate $A \rightarrow E$

$S \rightarrow a|aA|B|$

$A \rightarrow aBB$

$B \rightarrow Aa|b|a$

eliminate $S \rightarrow B$ (unit production)

$\bar{S} \rightarrow B$ $B \rightarrow Aa|b|a$

$S \rightarrow a|aA|Aa|b|a$

$A \rightarrow aBB$

$B \rightarrow Aa|b|a$

$A \rightarrow \text{min}$
 $\downarrow$ CFG

CNF productions:

$$S \rightarrow a$$

$$S \rightarrow b$$

$$B \rightarrow b$$

$$B \rightarrow a$$

1st It

$$S \rightarrow aA$$

$$S \rightarrow Aa$$

$$A \rightarrow aBB$$

$$B \rightarrow Aa$$

$$S \rightarrow (aA$$

$$S \rightarrow Aca$$

$$A \rightarrow (aBB$$

$$B \rightarrow Aca$$

$$(a \rightarrow a$$

$$(a \rightarrow a$$

$$(a \rightarrow B$$

$$(a \rightarrow a$$

✓

✓

x

$$A \rightarrow caBB$$

$$A \rightarrow (aD_1$$

2nd It

$$D_1 \rightarrow BB$$

✓

$$S \rightarrow caA$$

$$A \rightarrow (aD_1$$

$$S \rightarrow caA | Aca | a | b$$

$$A \rightarrow (aD_1$$

$$D_1 \rightarrow BB$$

$$B \rightarrow Aca | a | b$$

$$(a \rightarrow a$$

$$(b \rightarrow b$$

final grammar

Greibach Normal form (GNF)

$$\begin{matrix} A \rightarrow a \\ A \rightarrow av^* \end{matrix}$$

$$A \rightarrow a \alpha \rightarrow \text{one or more non terminals.}$$

NT

terminal

A CFG is of GNF if the prod are of the above form.

conversion of CFG to GNF

- 1) simplify grammar.
- 2) CNF (conversion)
- 3) Non terminals are to be replaced by $A_1(A_2, A_3, \dots)$

4) if prod is of form $A_i \rightarrow A_j B$

$$A_i \rightarrow A_j B$$

$i < j$ leave prod as it is.

⑤ $i > j$ perform substitution.

⑥ $i = j$ left recursion.

convert CFG to GNF

$$S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

first 2 conditions already satisfied.

3rd cond replace

with $S \rightarrow A_1$

& A with A_2

$$A_1 \rightarrow A_2 A_2 \mid 0$$

$$A_2 \rightarrow A_1 A_1 \mid 1$$

$$A_1 \rightarrow A_2 A_2 \mid 0 \rightarrow \textcircled{1}$$

$i=1, j=2$ ($1 < 2$) leave as it is.

$$A_2 \rightarrow A_1 A_1 \mid 1$$

$i=2, j=1$ ($2 > 1, i > j$) sub A_1 in A_2

$$A_2 \rightarrow A_2 A_1 A_1 \mid 0 A_1 \mid 1$$

$i=1, j=2 \rightarrow$ left recursion.

$$\frac{A_2 \rightarrow A_2 A_2 A_1}{A} \quad \frac{10A11}{B_1} \quad \frac{1}{B_2}$$

$$\boxed{\begin{array}{l} A \rightarrow B A' \\ A' \rightarrow \alpha A' \mid \epsilon \end{array}}$$

→ formula

$A' = B_2$
Here

$$A_2 \Rightarrow 0A1B2 \mid 1B2$$

$$B_2 \rightarrow A_2 A_1 B_2 \mid \epsilon \rightarrow \text{remove this } \epsilon \text{ prod'}$$

$$\checkmark A_2 \rightarrow 0A1B2 \mid 1B2 \mid 0A1 \rightarrow (2)$$

$$B_2 \rightarrow A_2 A_1 B_2 \mid A_2 A_1 \rightarrow (3)$$

Subs A_2 in A_1

$$A_1 \rightarrow A_2 A_2 \mid 0$$

$$\checkmark A_1 \rightarrow 0A1B2A2 \mid 1B2A2 \mid 0A1A2 \mid 1A2 \mid 0$$

now A_1 & A_2 are in GNF but

not B_2 so we sub A_2 in B_2

to convert B_2 to GNF

$$B_2 \rightarrow 0A1B2A2B2 \mid 0A1B2A1 \mid 1B2A1B2$$

$$1B2A1 \mid 0A1A1B2 \mid 0A1A1$$

$$1A1B2 \mid 1A2A1$$

20) $S \rightarrow (A) | BB$
 $B \rightarrow b | s B$
 $C \rightarrow b$
 $A \rightarrow a$

first 2 rules already satisfied.

- S with A_1
- B with A_2
- A with A_3
- B with A_4

$$A_1 \rightarrow A_2 A_3 | A_4 A_4$$

$$A_4 \rightarrow b | A_4 A_4$$

$$A_2 \rightarrow b$$

$$A_3 \rightarrow a$$

consider $A_1 \rightarrow A_2 A_3 | A_4 A_4$
 $i=1 \quad j=2$ leave prod now

consider $A_4 \rightarrow A_1 A_4 | b$
 $i=4 \quad j=1$ substitute on

$$A_4 \rightarrow A_2 A_3 A_4 | A_4 A_4 A_4 | b$$

$i=4 \quad j=2 \quad i > j$ substitute

$$A_4 \rightarrow b A_3 A_4 | A_4 A_4 A_4 | b$$

$i=4 \quad j=4$ left recursion

eliminate left recursion

$$\begin{aligned} A &\rightarrow A' \alpha | \beta \\ A &\rightarrow \beta A' \\ A' &\rightarrow \alpha A' | \epsilon \end{aligned}$$

So now,

$$\begin{array}{l} A_4 \rightarrow \underbrace{A_4 A_4 A_4}_{\alpha} \mid \underbrace{b A_3 A_4}_{\beta_1} \mid \underbrace{b}_{\beta_2} \end{array}$$

$$\begin{array}{l} A_4 \rightarrow b A_3 A_4 \mid b z_4 \\ z_4 \rightarrow A_4 A_4 z_4 \mid \epsilon \end{array}$$

eliminate $z_4 \rightarrow \epsilon$

In GNF, $A_4 \rightarrow b A_3 A_4 z_4 \mid b z_4 \mid b A_3 A_4 \mid b$

Not in GNF $z_4 \rightarrow A_4 A_4 z_4 \mid A_4 A_4$

convert z_4 to GNF, by sub A_4 in z_4

In GNF

$$\begin{array}{l} z_4 \rightarrow b A_3 A_4 z_4 A_4 z_4 \mid b A_3 A_4 z_4 A_4 \mid \\ \quad b z_4 A_4 z_4 \mid b z_4 A_4 \mid b A_3 A_4 A_4 z_4 \mid \\ \quad b A_3 A_4 A_4 \mid b A_4 z_4 \mid b A_4 \end{array}$$

consider $A_1 \rightarrow A_2 A_3 \mid A_4 A_4$

In GNF $\rightarrow$

$$\begin{array}{l} A_1 \rightarrow b A_3 \mid b A_3 A_4 z_4 A_4 \mid b z_4 A_4 \mid \\ \quad b A_3 A_4 A_4 \mid b A_4 \end{array}$$

Q)

$$A_1 \rightarrow A_2 A_3$$

$$A_2 \rightarrow A_3 A_1 \mid b$$

$$A_3 \rightarrow A_1 A_1 \mid a$$

first 3 rules satisfied,

consider $A_1 \rightarrow A_2 A_3$
 $i=1 \quad j=2 \quad (i < j)$ leave now

consider $A_2 \rightarrow A_3 A_1 | b$
 $i=2 \quad j=3 \quad (i < j)$ leave now

consider $A_3 \rightarrow A_1 A_2 | a$
 $i=3 \quad j=1 \quad (i > j)$ substitution

sub A_1 in A_3

$A_3 \rightarrow A_2 A_3 A_2 | a$

$i=3 \quad j=2$

$(i > j)$ substitution

$A_3 \rightarrow A_3 A_1 A_3 A_2 | \cancel{b} A_3 A_2 | a$

$i=3 \quad j=3 \quad (i=j)$ left recursion

consider,

$A_3 \rightarrow A_3 A_1 A_3 A_2 | b A_3 A_2 | a$

$A \rightarrow A$

$A_3 \rightarrow b A A_2 z_3 | a z_3$
 $z_3 \rightarrow A_1 A_3 A_2 z_3 | \epsilon$

Remove $z_3 \rightarrow \epsilon$

IN GNF
 NOT IN GNF

$A_3 \rightarrow b A A_2 z_3 | a z_3 | b A A_2 | a$
 $z_3 \rightarrow A_1 A_3 A_2 z_3 | A_1 A_3 A_2$

sub A_3 in A_2 ,

$$\text{GNF} \left\{ \begin{array}{l} A_2 \rightarrow bA_3A_2Z_3A_1 \mid aZ_3A_1 \mid bA_3A_2A_1 \mid aA_1 \mid b \end{array} \right.$$

sub A_2 , val in A_1

$$A_1 \rightarrow A_2A_3$$

$$\left\{ \begin{array}{l} A_1 \rightarrow bA_3A_2Z_3A_1A_3 \mid aZ_3A_1A_3 \mid bA_3A_2A_1A_3 \mid aA_1A_3 \mid bA_1 \end{array} \right.$$

is in GNF.

Now sub A_1 in Z_3 ,

$$Z_3 \rightarrow bA_3A_2Z_3A_1A_3A_3A_2Z_3 \mid bA_3A_2Z_3A_1A_3A_3A_2 \mid$$

$$aZ_3A_1A_3A_3A_2Z_3 \mid aZ_3A_1A_3A_3A_2 \mid$$

$$bA_3A_2A_1A_3A_3A_2Z_3 \mid bA_3A_2A_1A_3A_3A_2 \mid$$

$$aA_1A_3A_3A_2A_3 \mid aA_1A_3A_3A_2 \mid bA_3A_3A_2A_3 \mid$$

$$bA_3A_3A_2$$

pumping lemma for CFL

pumping lemma is used to prove a language is not context free.

let L be context free, let n be a integer constant select a string z such

that $|z| \geq n$.

Divide the string z into 5 parts $uvwx^iy$

such $|vwx| \leq n$, $|vx| \geq 1$ for $i \geq 0$

uv^iwx^iy is L

show that $L = \{ a^n b^n \mid n \geq 1 \}$ is not CFL

let L be a CFL let $n=3$

$$L = \{ a^3 b^3, a^4 b^4, a^5 b^5, \dots \}$$

$$z = a^3 b^3 \quad |z| \geq n$$

$$|a^3 b^3| \geq 3$$

$$\begin{array}{c} a^3 b^3 \\ \hline u \quad v \quad w \quad x \quad y \end{array} \quad \begin{array}{l} u=aa \quad v=a \\ w=b \quad x=b \quad y=bbb \end{array}$$

$$|uwx| \leq n$$

$$|vx| \geq 1 \Rightarrow |ab| \leq 3$$

$$|a| \geq 1 \quad 3 \leq 3$$

$$uv^iwx^iy$$

$$i=0, \quad a^3 b^3 \in L$$

$$a^3 b^3 \notin L \text{ not CFL}$$

Q) prove that $L = \{ a^p \mid p \text{ is prime number} \}$ is not CFL

let L be CFL

$$L = \{ aa, aaa, aaaaa, aaaaaa, \dots \}$$

$$n=5$$

$$z = aaaaa$$

$$|z| \geq n$$

$$|aaaaa| \geq 5$$

$$5 \geq 5$$

$$i=3,$$

$$a^3 a^3 a$$

$$= a^9 \notin L$$

contradiction and L not CFL.

$$z = aaaaa$$

$$|uwx| \leq n$$

$$|vx| \geq 1$$

$$uv^iwx^iy, i=0$$

$$a^3 a^3 a = a^9 \in L$$

$$i=1$$

$$aaaaa \in L$$

$$i=2$$

$$aaaaaa$$

Q) prove that $L = \{a^n \mid n \geq 1\}$ is not a CFL.
 $L = \{a, aaaa, aaaaaaaaaa, \dots\}$

let $L \in CFL$ $z = a^4$

let $n = 2$

$z = aaaa$

$|vwx| \leq 4$

$2 \leq 4$

$|z| \geq n$

$4 \geq 2$

$|vwx| \geq 1$

$2 \geq 1$

uv^iwx^iy

$i=0$ $aa^0 \in a^0a$

$aa \notin L$ contradiction.

L is not a CFL.

Q) show that $L = \{a^i b^j \mid j = i^2\}$ is not a CFL.

$L = \{ab, aabbbb, aaaa b b b b b b b b \dots\}$

let $n = 4$ $z = aabbbb$

$|z| \geq n$

$aa b b b b$

$\overline{u} \overline{v} \overline{w} \overline{x} \overline{y}$

$|vwx| \leq 4$

$4 \leq 4$

$|vwx| \geq 1$

$(ab)^2$

$2 \geq 1$

uv^iwx^iy

$i=0$

$aa^0 b b^0 b b^0 b b^0$

$abbb \notin L$ contradiction

L is not CFL.

diff b/w regular lang and context free lang

Regular lang

- * language which is generated by regular grammar.
- * regular lang is accepted by FA such as DFA & NFA.
- * regular languages are closed under union, concat, reverse, closure, Intersection, complement
- * regular lang are used in sequential circuit & text editors.

Context free lang

- * language generated by CFG.
- * CFL is accepted by PDA.
- * CFL's are closed under union, concat, reverse closure
- * CFL's are used in compiler.

closure properties of CFL

CFL's are closed under,

- a) union b) concatenation c) power closure.

Not closed under

- a) Intersection b) complementation.

union! - let L_1 & L_2 CFL's

$L_1 \cup L_2$ is also a CFL.

$$L_1 = \{a^n \mid n \geq 0\}$$

$$L_2 = \{b^n \mid n \geq 0\}$$

$$L_1 = \{\epsilon, a, aa, \dots\}$$

$$L_2 = \{\epsilon, b, bb, \dots\}$$

1) $a \in L_1$

$$2) \epsilon \in L_2, L_3 = L_1 \cup L_2 = \{\epsilon, a, aa, b, bb, \dots\}$$

$\{S \rightarrow S_1 \mid S_2\} \rightarrow$ so $L_1 \cup L_2$ also CFL.

b) concatenation,

$$L_1 \cdot L_2$$

$$L_1 = \{a^n b^m\}$$

$$L_2 = \{b^n a^m\}$$

$$S_2 \rightarrow b S_2 | \epsilon$$

$$L_1 \cdot L_2 = L_3 \Rightarrow$$

$$S \rightarrow S_1 S_2$$

c) Pumping lemma,

L is a CFL

L^* is also a CFL.

$$L = \{a\}$$

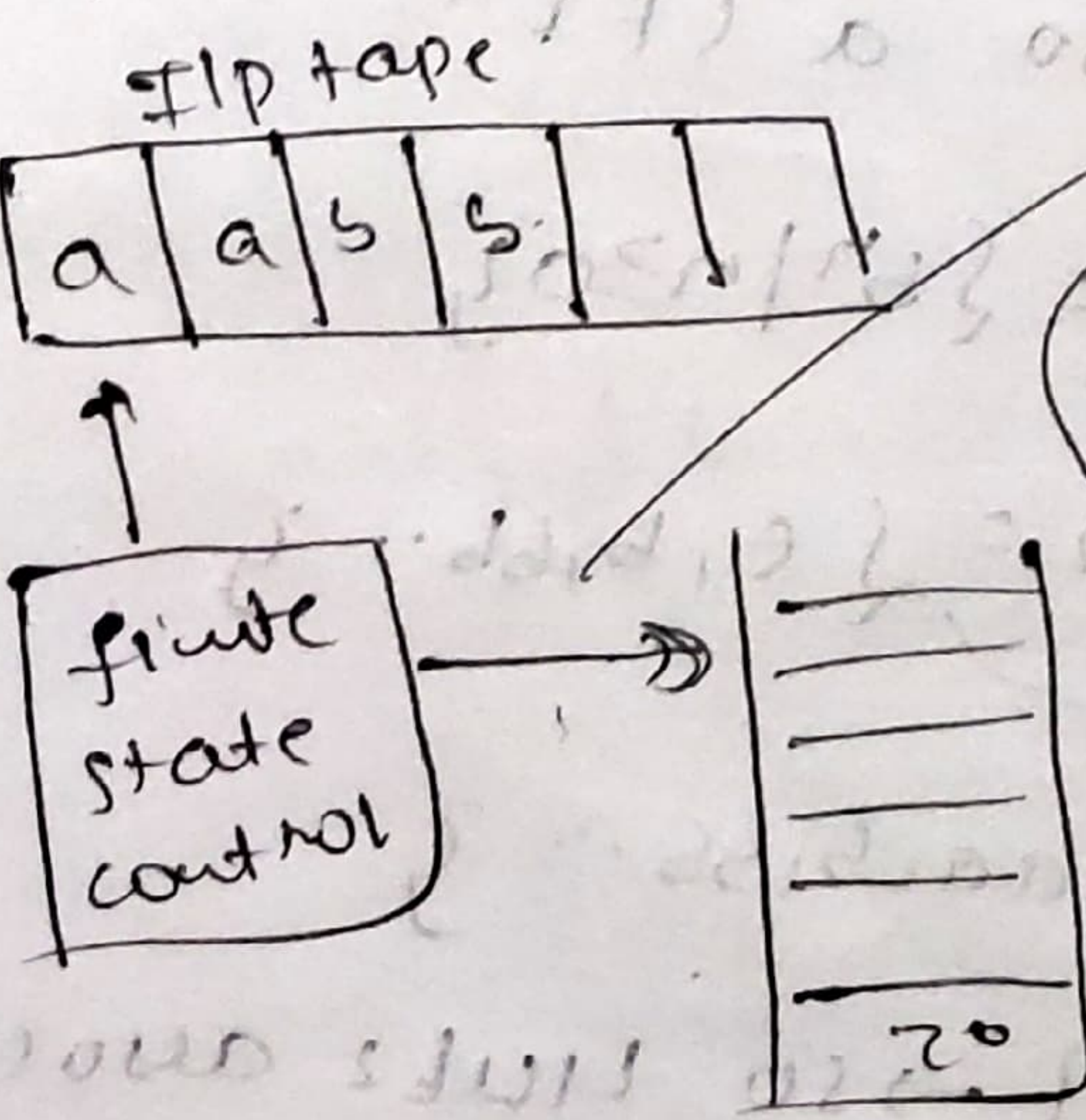
$$L^* = \{\epsilon, a, aa, \dots\}$$

Push Down Automata (PDA)

FA has limited memory and cannot handle complex languages.

PDA $\rightarrow$ accepts CFL / recognized by PDA.

Model of PDA.



always points to top of stack.

push pop

stack

$\delta(q_0, a) \rightarrow FA$
 $\delta(q_0, a, b)$
 PDA

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

Q is finite set of states

Σ is an input alphabet

Γ is an stack alphabet

q_0 is initial state

z_0 is start/top symbol of stack

F is set of final states

δ is transition function which maps

$$\text{from } Q \times (\Sigma \cup \{\epsilon\}) \times \Gamma \rightarrow Q \times \Gamma^*$$

$\Gamma^* \rightarrow$ a string of symbol that are replaced in the stack

Q) construct PDA for language $L = \{a^n b^n \mid n \geq 1\}$

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

$$L = \{ab, aabb, aaabbb, \dots\}$$

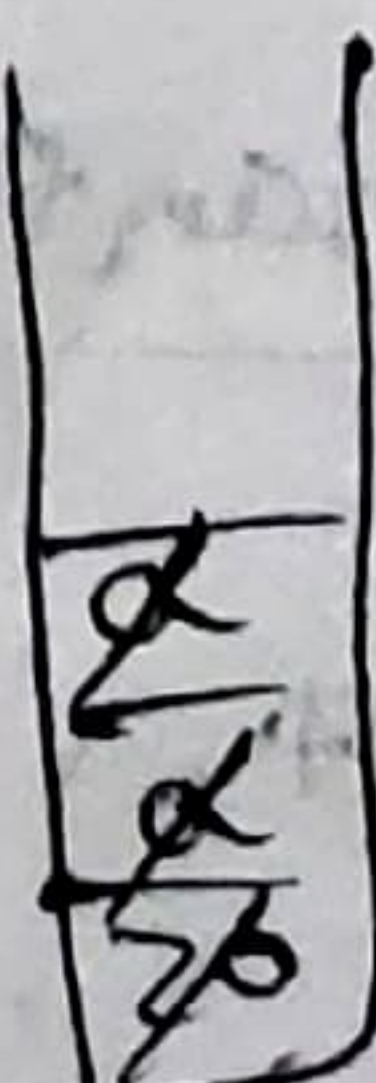
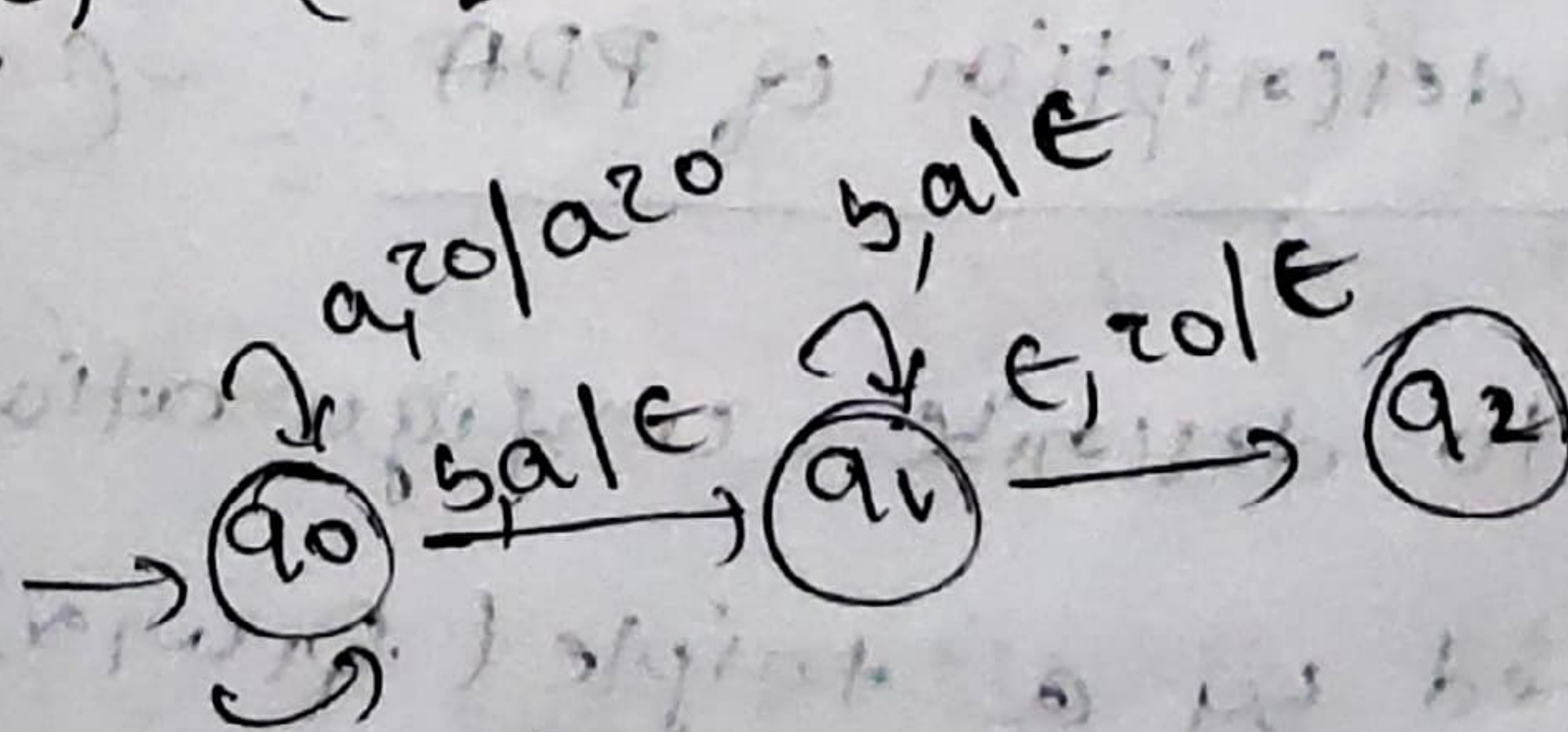
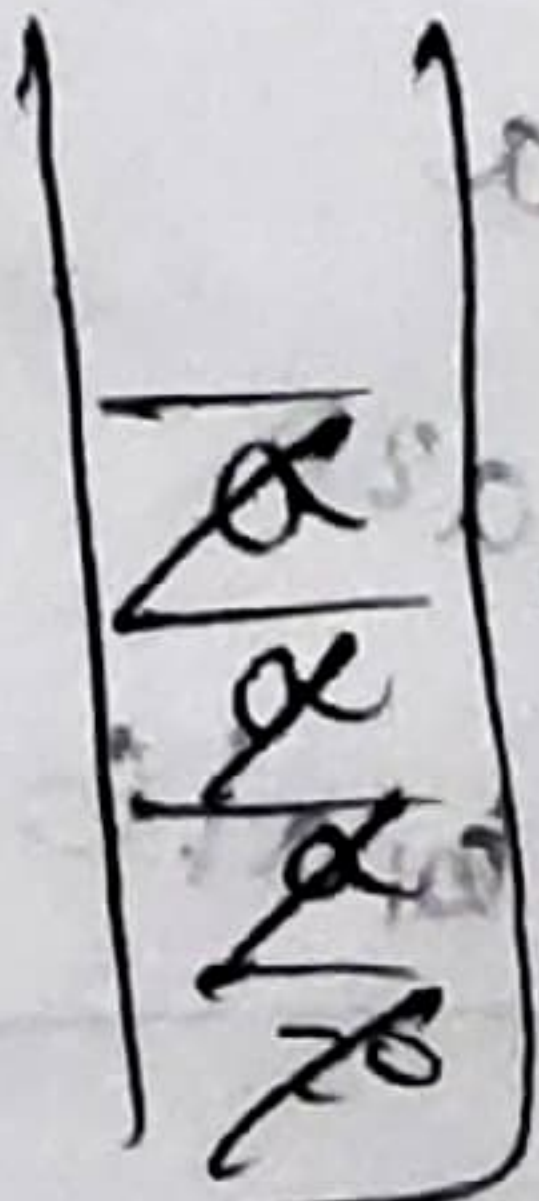
$$\delta(q_0, a, z_0) = \delta(q_0, a, z_0) \quad \boxed{a \mid a \mid a \mid b \mid b \mid b \mid \epsilon}$$

$$\delta(q_0, a, a) = \delta(q_0, a, a, z_0)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$



aaab.

$$(q_0, aab, z_0) \vdash (q_0, abb, az_0) \vdash (q_0, b, aaz_0) \vdash (q_1, \epsilon, z_0) \vdash (q_2, \epsilon)$$

$$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{z_0, a\}, \delta, q_0, z_0, \{q_2\})$$

graphical notation for PDA :

Draw graphical notation for following PDA transitions.

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

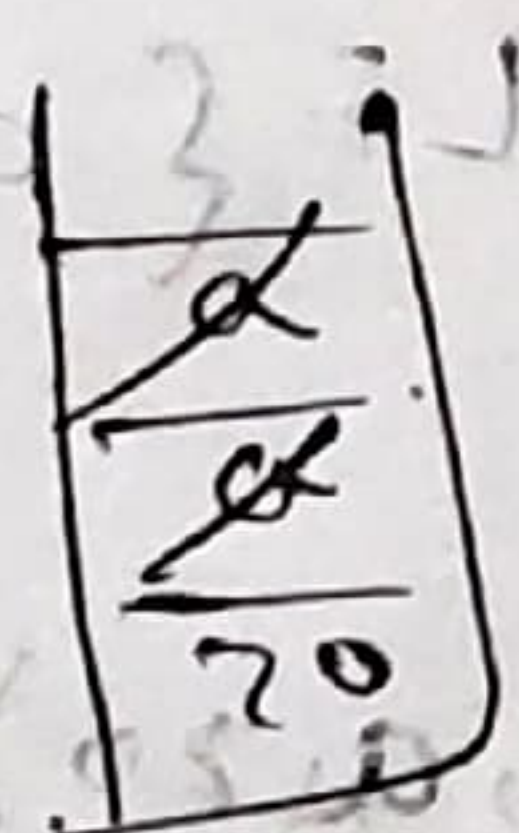
$$\delta(q_0, a, z_0) = (q_0, az_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

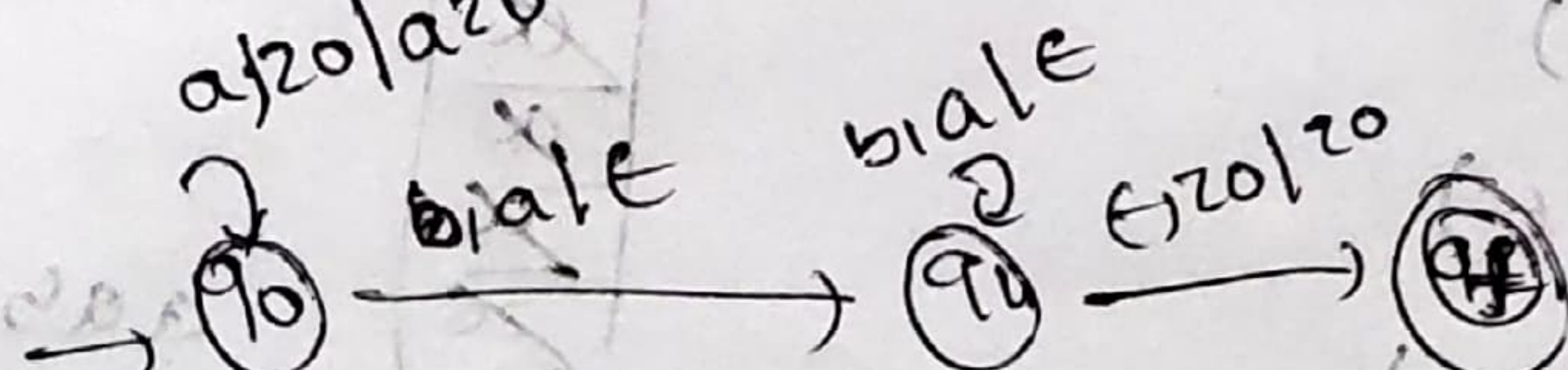
$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, z_0)$$



a/a

az/az



Instantaneous description of PDA :- (ID)

* ID is used to describe configuration of PDA.

* ID is defined by a triple (q, w, γ)

* q means current state, w means string, γ -> content of stack

ex: $L = \{a^n b^n \mid n \geq 1\}$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, a, a) = (q_0, a a z_0)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$

$$\begin{aligned} (q_0, a a b b, z_0) &\vdash (q_0, a a b b, a z_0) \vdash (q_0, b b, a a z_0) \vdash \\ (q_1, b, a z_0) &\vdash (q_1, \epsilon, z_0) \vdash (q_2, \epsilon) \text{ final state} \end{aligned}$$

Acceptance of PDA:

1) Acceptance by final state.

2) " " empty stack.

$$(q_1, w, z_0) \xrightarrow{*} (q_2, \epsilon) \text{ where } q_2 \in F$$

$$(q_1, w, z_0) \xrightarrow{*} (p, \epsilon) \text{ where } p \in Q$$

$$\begin{aligned} p &\in Q \\ \forall \epsilon &\in \epsilon^* \end{aligned}$$

$L = \{a^n b^n \mid n \geq 1\}$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, a, a) = (q_0, a a z_0)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon) \rightarrow \text{empty stack}$$

$$(q_2, z_0) \rightarrow \text{final state}$$

$$(q_0, a a b b, z_0) \vdash (q_0, a a b b, a z_0) \vdash (q_0, b b, a a z_0) \vdash$$

$$(q_1, b, a z_0) \vdash (q_1, \epsilon, z_0) \vdash (q_2, z_0) \text{ or } (q_1, \epsilon) \text{ stack empty}$$

* After processing entire string if we reach to final state then we say that string/lang accepted by PDA.

Q) Construct PDA for language $L = \{a^n b^n \mid n \geq 1\}$

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

$$L = \{abb, aabbbb, \dots\}$$

a | a | b | b | b | b | ε

Logic: if $a \rightarrow$ push 2 a's : if $a \rightarrow$ push a
 $b \rightarrow$ pop : for even no b
 pop 'a'

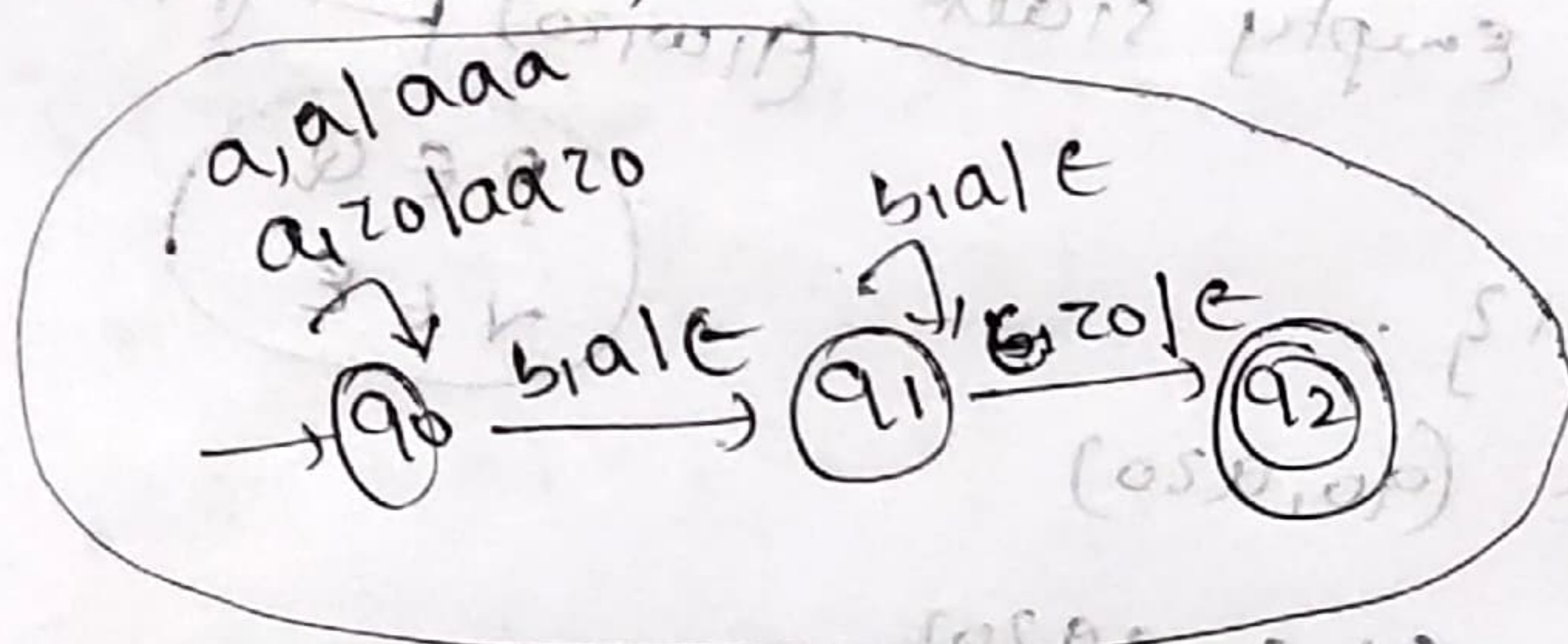
$$\delta(q_0, a, z_0) = \epsilon(q_0, aaz_0)$$

$$\delta(q_0, a, a) = (q_0, aaaa)$$

$$\delta(q_0, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, a) = (q_1, \epsilon)$$

$$\delta(q_1, \epsilon, z_0) = (q_2, \epsilon)$$



$$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, z_0\}, \delta, q_0, z_0, \{q_2\})$$

Q) Construct PDA for $L = \{a^n b^n \mid n \geq 1\}$

$$L = \{abb, aabbb, aaaaaabbb, \dots\}$$

Logic: 1) push first 'a' onto stack.

2) Ignore next a

3) Repeat step 1 & step 2 until all a's are covered.

4) when 'b' comes pop stack.

a | a | a | a | b | b | ε

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

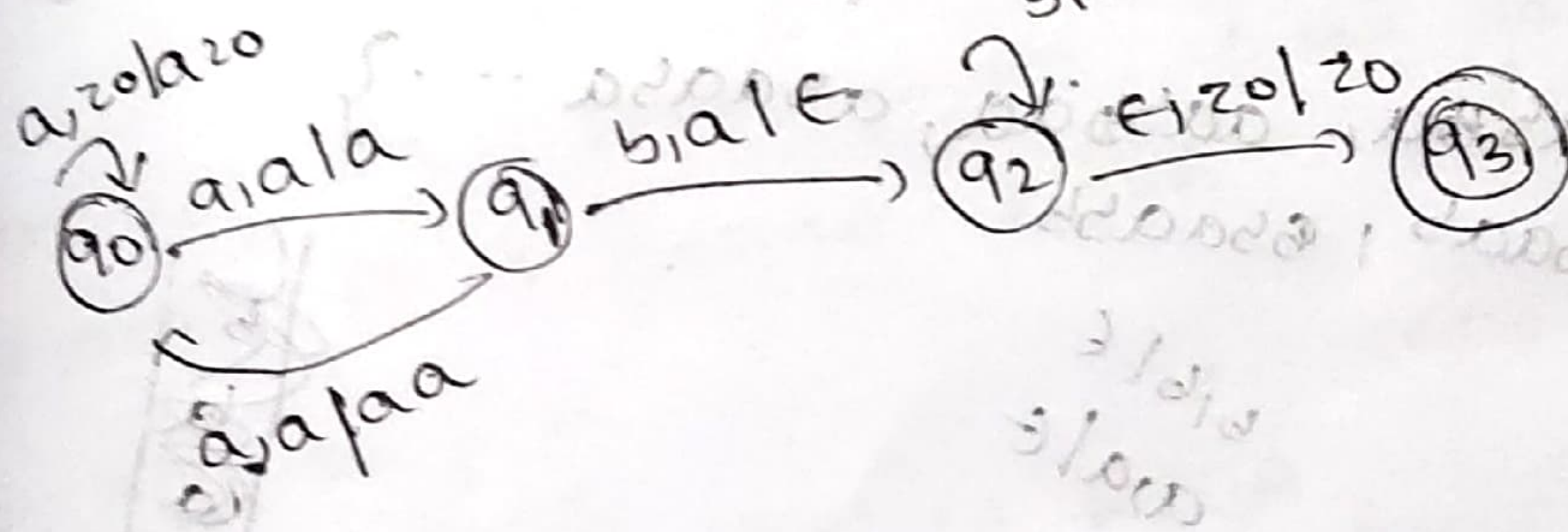
$$\delta(q_0, a, a) = (q_1, a)$$

$$\delta(q_1, a, a) = (q_0, aa)$$

$$\delta(q_1, b, a) = \delta(q_2, \epsilon)$$

$$\delta(q_2, b, a) = \delta(q_2, \epsilon)$$

$$\delta(q_2, \epsilon, z_0) = \delta(q_3, \epsilon) \text{ or } \delta(q_3, z_0)$$



$$M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, z_0\}, \delta, q_0, z_0, \{q_3\})$$

$$L = \{w | w \in (a+b)^*\}$$

$$L = \{a^5 a c a b a, a b a b c b a b a, \dots\}$$

$$\delta(q_0, a, z_0) = (q_0, a z_0)$$

$$\delta(q_0, b, z_0) = (q_0, b z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, a, b) = (q_0, ab)$$

$$\delta(q_0, b, a) = (q_0, ba)$$

$$\delta(q_0, b, b) = (q_0, bb)$$

$$\delta(q_0, c, a) = (q_1, a)$$

$$\delta(q_0, c, b) = (q_1, b)$$

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

$$\delta(q_1, b, b) = (q_1, \epsilon)$$

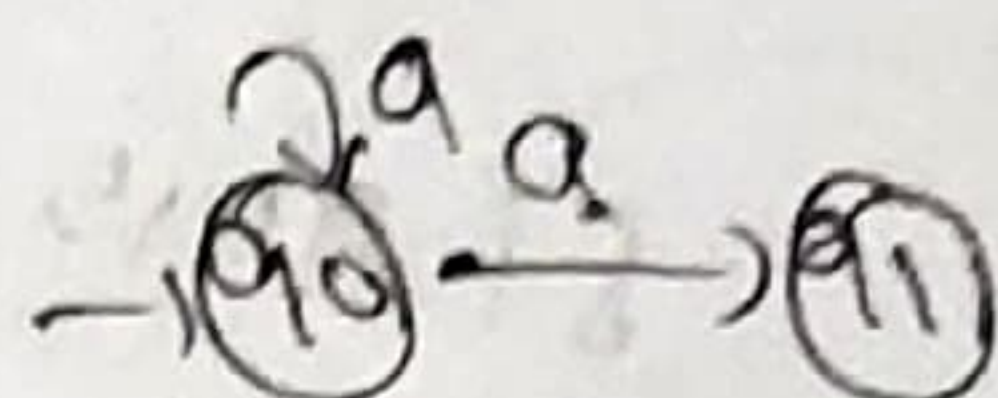
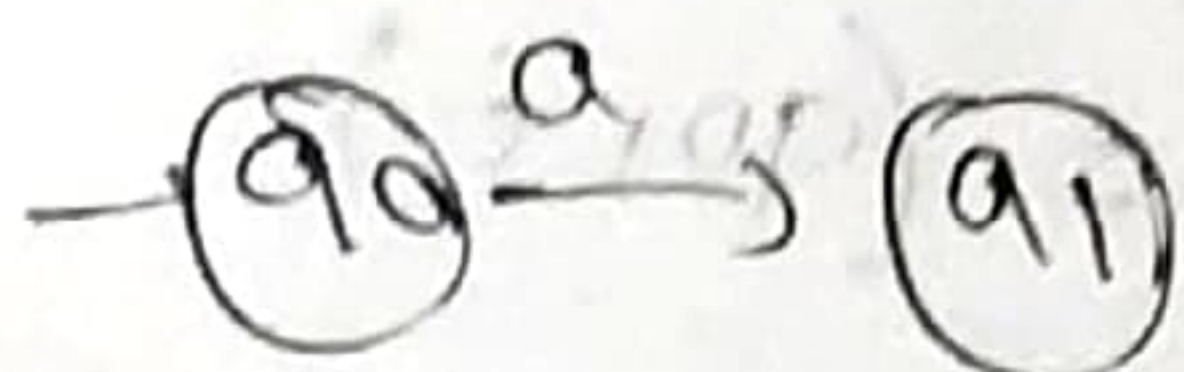
$$\delta(q_1, \epsilon, z_0) = \delta(q_2, \epsilon)$$

a
b
a
z0

Deterministic push down Automata (DPDA)



DPDA



PPDA : 105, 130, 150, 170, 190, 210, 230, 250, 270, 290, 310, 330, 350, 370, 390, 410, 430, 450, 470, 490, 510, 530, 550, 570, 590, 610, 630, 650, 670, 690, 710, 730, 750, 770, 790, 810, 830, 850, 870, 890, 910, 930, 950, 970, 990, 1010, 1030, 1050, 1070, 1090, 1110, 1130, 1150, 1170, 1190, 1210, 1230, 1250, 1270, 1290, 1310, 1330, 1350, 1370, 1390, 1410, 1430, 1450, 1470, 1490, 1510, 1530, 1550, 1570, 1590, 1610, 1630, 1650, 1670, 1690, 1710, 1730, 1750, 1770, 1790, 1810, 1830, 1850, 1870, 1890, 1910, 1930, 1950, 1970, 1990, 2010, 2030, 2050, 2070, 2090, 2110, 2130, 2150, 2170, 2190, 2210, 2230, 2250, 2270, 2290, 2310, 2330, 2350, 2370, 2390, 2410, 2430, 2450, 2470, 2490, 2510, 2530, 2550, 2570, 2590, 2610, 2630, 2650, 2670, 2690, 2710, 2730, 2750, 2770, 2790, 2810, 2830, 2850, 2870, 2890, 2910, 2930, 2950, 2970, 2990, 3010, 3030, 3050, 3070, 3090, 3110, 3130, 3150, 3170, 3190, 3210, 3230, 3250, 3270, 3290, 3310, 3330, 3350, 3370, 3390, 3410, 3430, 3450, 3470, 3490, 3510, 3530, 3550, 3570, 3590, 3610, 3630, 3650, 3670, 3690, 3710, 3730, 3750, 3770, 3790, 3810, 3830, 3850, 3870, 3890, 3910, 3930, 3950, 3970, 3990, 4010, 4030, 4050, 4070, 4090, 4110, 4130, 4150, 4170, 4190, 4210, 4230, 4250, 4270, 4290, 4310, 4330, 4350, 4370, 4390, 4410, 4430, 4450, 4470, 4490, 4510, 4530, 4550, 4570, 4590, 4610, 4630, 4650, 4670, 4690, 4710, 4730, 4750, 4770, 4790, 4810, 4830, 4850, 4870, 4890, 4910, 4930, 4950, 4970, 4990, 5010, 5030, 5050, 5070, 5090, 5110, 5130, 5150, 5170, 5190, 5210, 5230, 5250, 5270, 5290, 5310, 5330, 5350, 5370, 5390, 5410, 5430, 5450, 5470, 5490, 5510, 5530, 5550, 5570, 5590, 5610, 5630, 5650, 5670, 5690, 5710, 5730, 5750, 5770, 5790, 5810, 5830, 5850, 5870, 5890, 5910, 5930, 5950, 5970, 5990, 6010, 6030, 6050, 6070, 6090, 6110, 6130, 6150, 6170, 6190, 6210, 6230, 6250, 6270, 6290, 6310, 6330, 6350, 6370, 6390, 6410, 6430, 6450, 6470, 6490, 6510, 6530, 6550, 6570, 6590, 6610, 6630, 6650, 6670, 6690, 6710, 6730, 6750, 6770, 6790, 6810, 6830, 6850, 6870, 6890, 6910, 6930, 6950, 6970, 6990, 7010, 7030, 7050, 7070, 7090, 7110, 7130, 7150, 7170, 7190, 7210, 7230, 7250, 7270, 7290, 7310, 7330, 7350, 7370, 7390, 7410, 7430, 7450, 7470, 7490, 7510, 7530, 7550, 7570, 7590, 7610, 7630, 7650, 7670, 7690, 7710, 7730, 7750, 7770, 7790, 7810, 7830, 7850, 7870, 7890, 7910, 7930, 7950, 7970, 7990, 8010, 8030, 8050, 8070, 8090, 8110, 8130, 8150, 8170, 8190, 8210, 8230, 8250, 8270, 8290, 8310, 8330, 8350, 8370, 8390, 8410, 8430, 8450, 8470, 8490, 8510, 8530, 8550, 8570, 8590, 8610, 8630, 8650, 8670, 8690, 8710, 8730, 8750, 8770, 8790, 8810, 8830, 8850, 8870, 8890, 8910, 8930, 8950, 8970, 8990, 9010, 9030, 9050, 9070, 9090, 9110, 9130, 9150, 9170, 9190, 9210, 9230, 9250, 9270, 9290, 9310, 9330, 9350, 9370, 9390, 9410, 9430, 9450, 9470, 9490, 9510, 9530, 9550, 9570, 9590, 9610, 9630, 9650, 9670, 9690, 9710, 9730, 9750, 9770, 9790, 9810, 9830, 9850, 9870, 9890, 9910, 9930, 9950, 9970, 9990, 10010, 10030, 10050, 10070, 10090, 10110, 10130, 10150, 10170, 10190, 10210, 10230, 10250, 10270, 10290, 10310, 10330, 10350, 10370, 10390, 10410, 10430, 10450, 10470, 10490, 10510, 10530, 10550, 10570, 10590, 10610, 10630, 10650, 10670, 10690, 10710, 10730, 10750, 10770, 10790, 10810, 10830, 10850, 10870, 10890, 10910, 10930, 10950, 10970, 10990, 11010, 11030, 11050, 11070, 11090, 11110, 11130, 11150, 11170, 11190, 11210, 11230, 11250, 11270, 11290, 11310, 11330, 11350, 11370, 11390, 11410, 11430, 11450, 11470, 11490, 11510, 11530, 11550, 11570, 11590, 11610, 11630, 11650, 11670, 11690, 11710, 11730, 11750, 11770, 11790, 11810, 11830, 11850, 11870, 11890, 11910, 11930, 11950, 11970, 11990, 12010, 12030, 12050, 12070, 12090, 12110, 12130, 12150, 12170, 12190, 12210, 12230, 12250, 12270, 12290, 12310, 12330, 12350, 12370, 12390, 12410, 12430, 12450, 12470, 12490, 12510, 12530, 12550, 12570, 12590, 12610, 12630, 12650, 12670, 12690, 12710, 12730, 12750, 12770, 12790, 12810, 12830, 12850, 12870, 12890, 12910, 12930, 12950, 12970, 12990, 13010, 13030, 13050, 13070, 13090, 13110, 13130, 13150, 13170, 13190, 13210, 13230, 13250, 13270, 13290, 13310, 133

925

$$L = \{ a^n b^m \mid n \geq 1 \}$$

$$L = \{ a^n b^{2n} \mid n \geq 1 \}.$$

$$L = \{ w c w^R \mid w \in (a+b)^* \} \cup L = \{ w \mid n_a(w) = n_b(w) \}$$

NPDA $\rightarrow \{ w w^R \mid w \in (a+b)^* \}$

for app or ILP symbol if we go to only one state it is called DPDA.

state it is called NPDA
* on app any one state it is called NPDA

a) construct PDA for $L = \{w \mid n_a(w) = n_b(w)\}$

$$L = \{ \epsilon, (ab)^n, a^n b^n, a^n a^n b^n, a^n b^n a^n, b^n a^n b^n, b^n b^n a^n \}$$

1) Empty

1) Empty $a \rightarrow \text{push}$

2) top $a \rightarrow \text{IIP}$ $a \rightarrow \text{pop}$

3) top $\rightarrow$ 5 $\rightarrow$ push 5
 top $\rightarrow$ 4 $\rightarrow$ pop

$$g(q_0, a, z_0) = (q_0, a, z_0)$$

$$\delta(q_0, b_1 z_0) = (q_0, b_2 z_0)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$S(90, 5, 6) = (90, 55)$$



$$S(90, a, b) = (90, e)$$

$$S(90, 1, a), f_0(90, t).$$

$$S(90, t, 20) = (91, t)$$

6120/620

isolato

$\text{90} \xrightarrow{E_{120/20}} \text{91}$

an/a/an

516153

215/15

5121

PDA for Balanced parenthesis

() 43 ()

$$((234))$$

(23)

$$g(90, 0, 20) = (90, 0, 20)$$

$$g(90, \{, 20) = (90, \{ 20)$$

$$g(90, c, c) = (90, cc)$$

$$S(90, 212) = (90, 212)$$

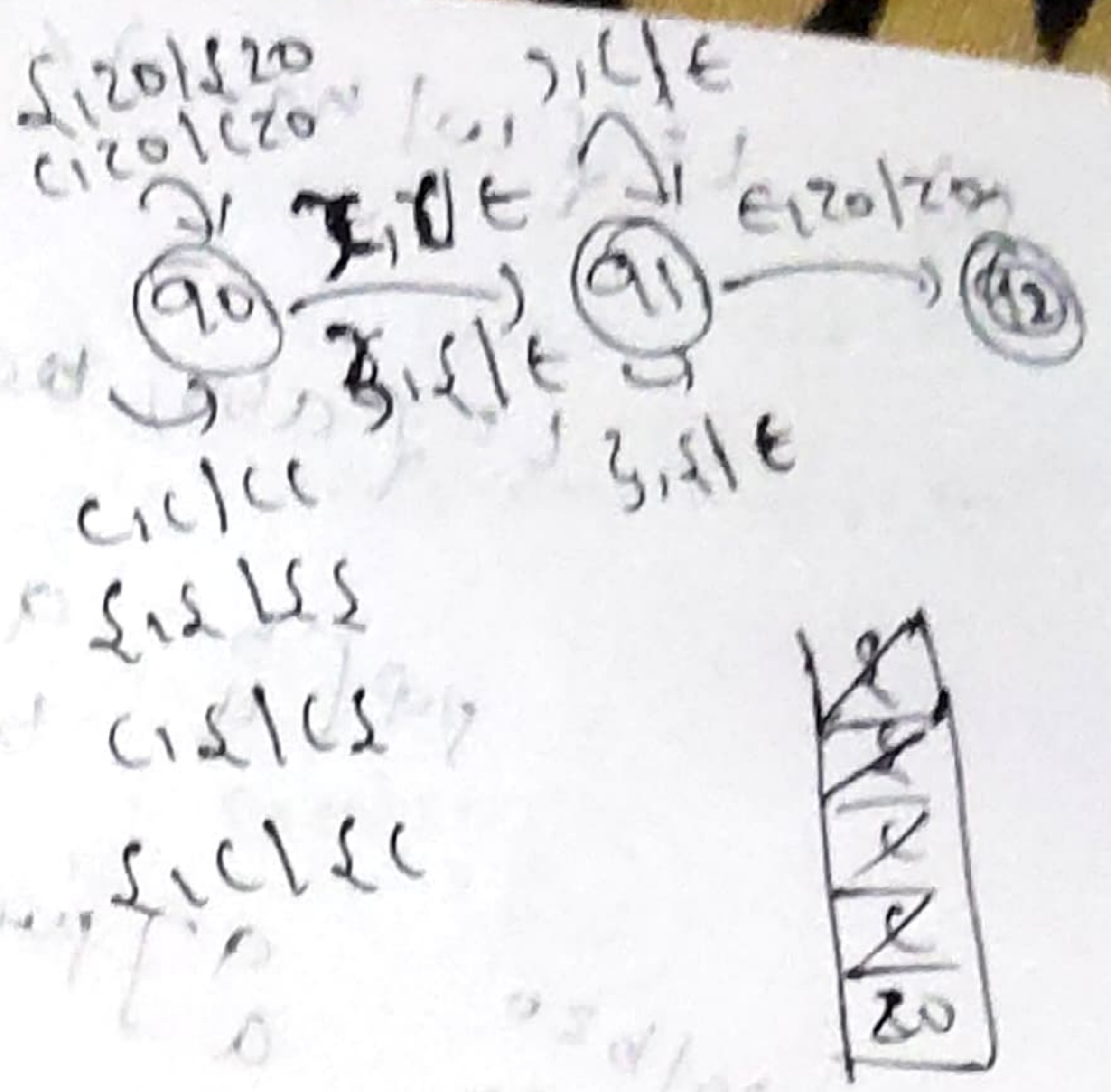
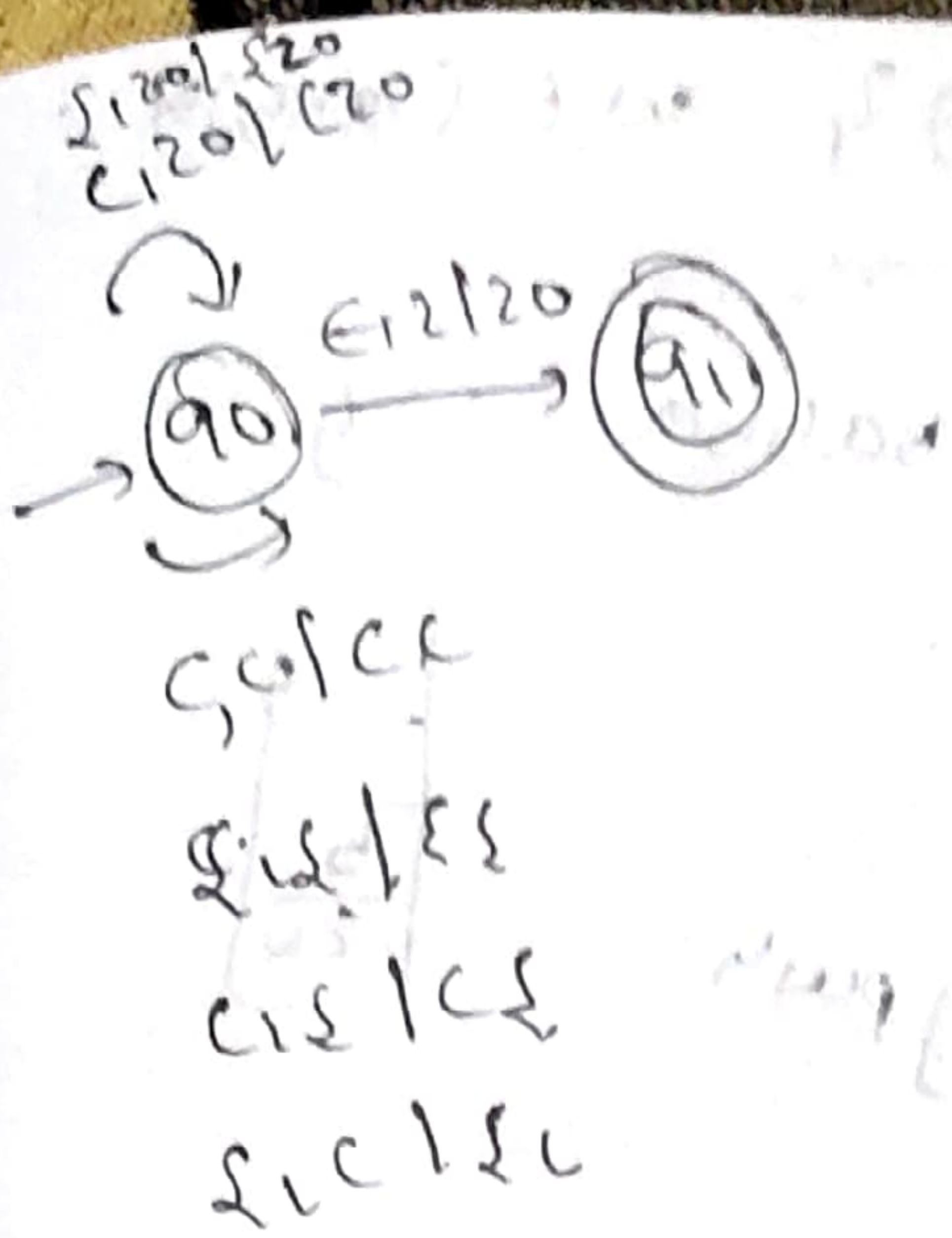
$$g(q_0, \gamma, c) = (q_0, t)$$

$$f(90, y, z) = (90, e)$$

$$\delta(a_0, f, c) = \{a_0, f(c)\}$$

$$\zeta(q_0, c, \ell) = (q_0, c, \ell)$$

$$\delta(a_0, e, z_0) = (a_1, e)$$

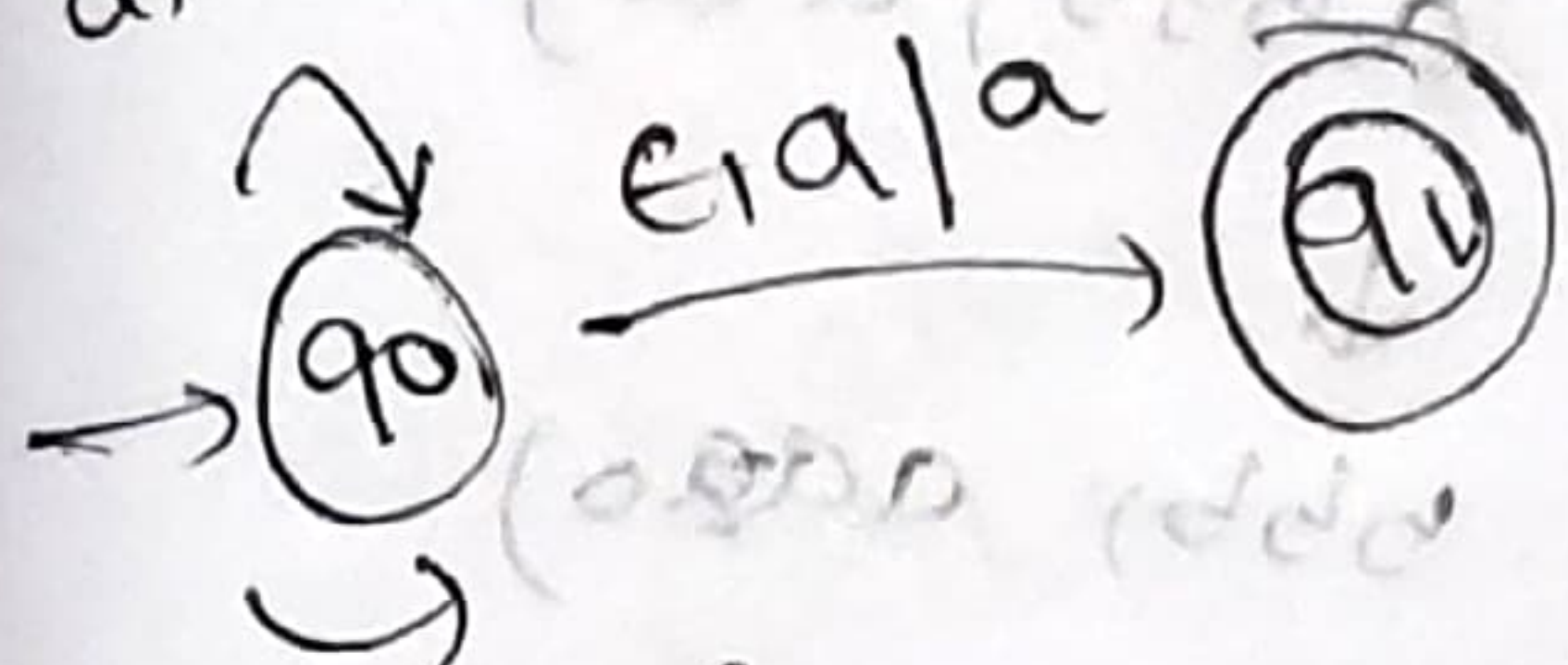


2
2
2
20

Q1 $L = \{w \mid n_a(w) > n_b(w) \text{ where } w \in (a+b)^*\}$

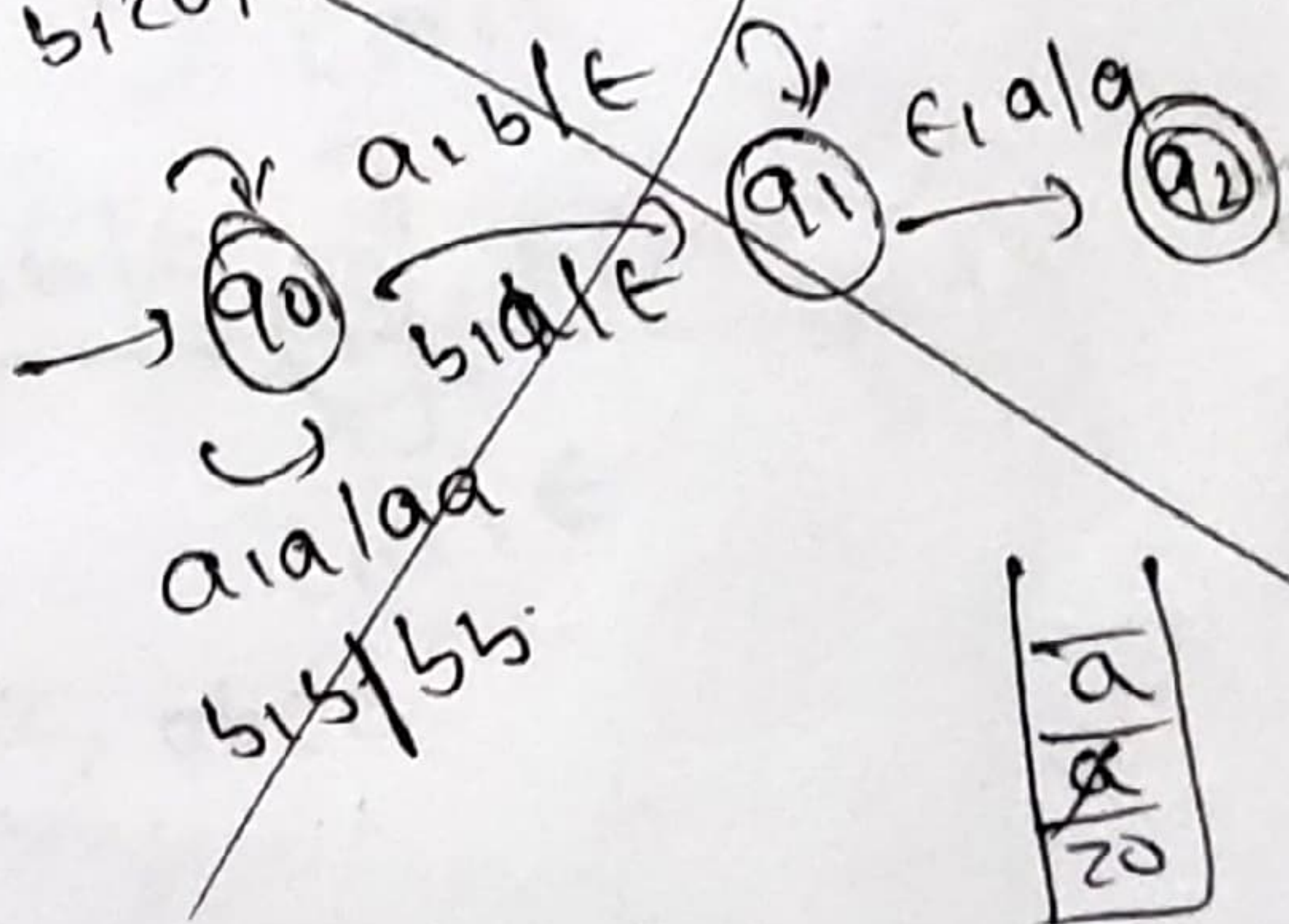
$L = \{aas, baabaa, asasaa, \dots\}$

b_1z_0/b_2z_0
 a_1z_0/a_2z_0



$a, aaaa$
 b, bbb
 a, bbb
 $b, aaaa$

a_1z_0/a_2z_0
 b_1z_0/b_2z_0



2
2
2
20

$(q_0, baaba, z_0)$

$(q_0, aasa, b_2z_0)$

(q_0, asa, z_0)

(q_0, ba, a_2z_0)

(q_0, a, z_0)

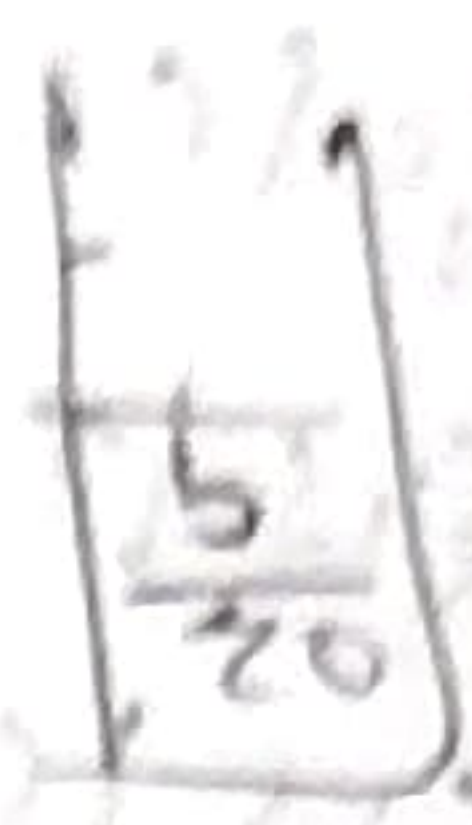
(q_0, ϵ, a_2z_0)

(q_1, ϵ, a_2z_0)

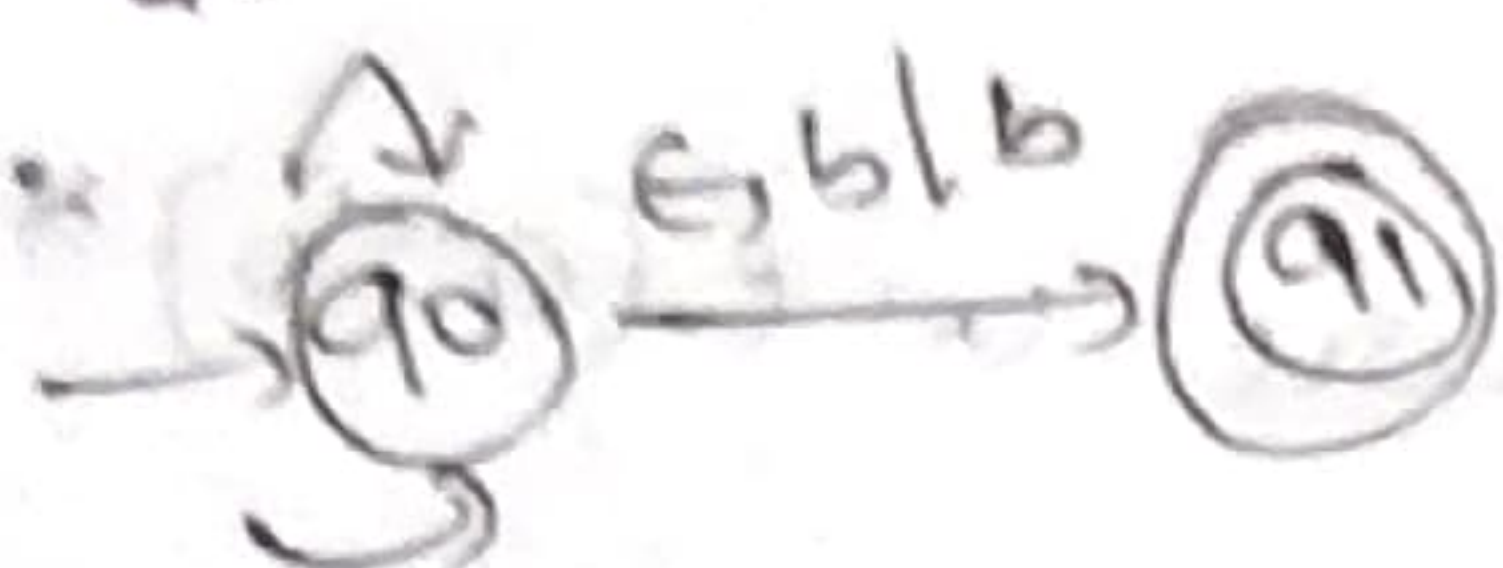
Q) $L = \{ w \mid \text{number of } a's \text{ in } w \text{ is even} \}$
 $L = \{ \epsilon, ab, ba, aabb, babb, \dots \}$

pop. [b a] pop

a] push b] push



b, 20 / b, 20
a, 20 / a, 20



$(q_0, a, 20)$

$(q_0, ba, 20)$

$(q_0, aabb, 20)$

$(q_0, abbb, 20)$

$(q_0, bbb, 20)$

$(q_0, bb, 20)$

$(q_0, b, 20)$

$(q_0, \epsilon, 20)$

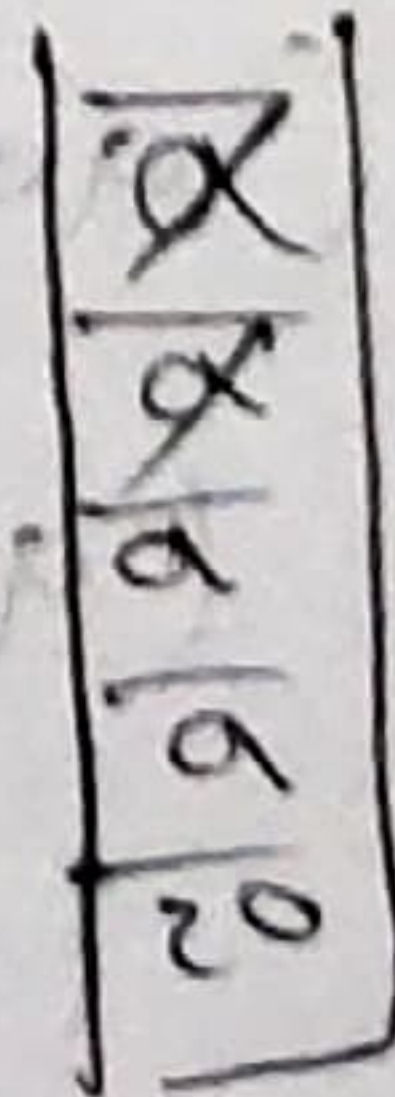
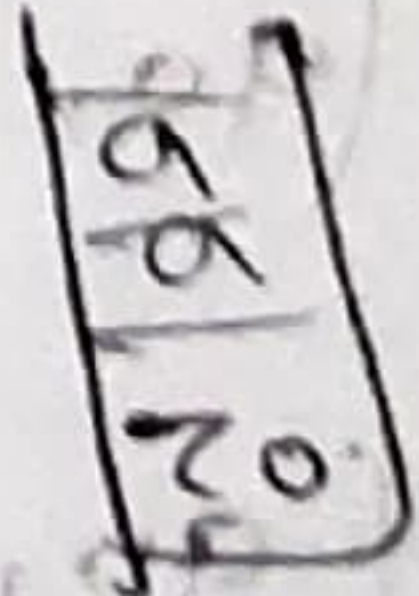
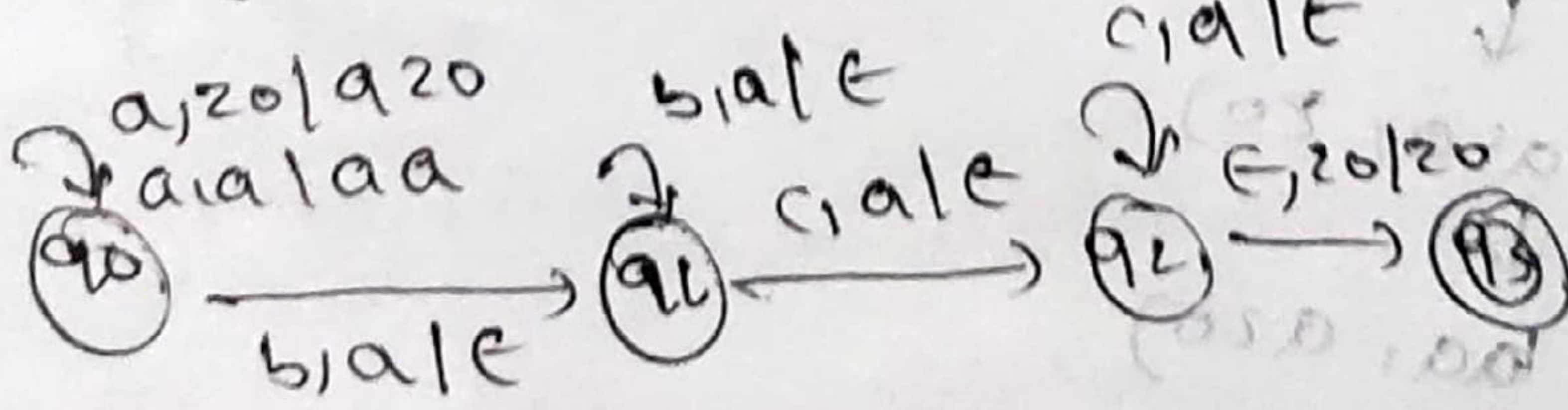
$(q_1, \epsilon, 20)$

b, 20 / b, 20
a, 20 / a, 20
b, 20 / b, 20
a, 20 / a, 20

ba, 20 / ba, 20

Q) $L = \{ a^m b^n c^p \mid m, n, p \geq 1 \}$

$L = \{ aabb, aaabbbcc, \dots \}$

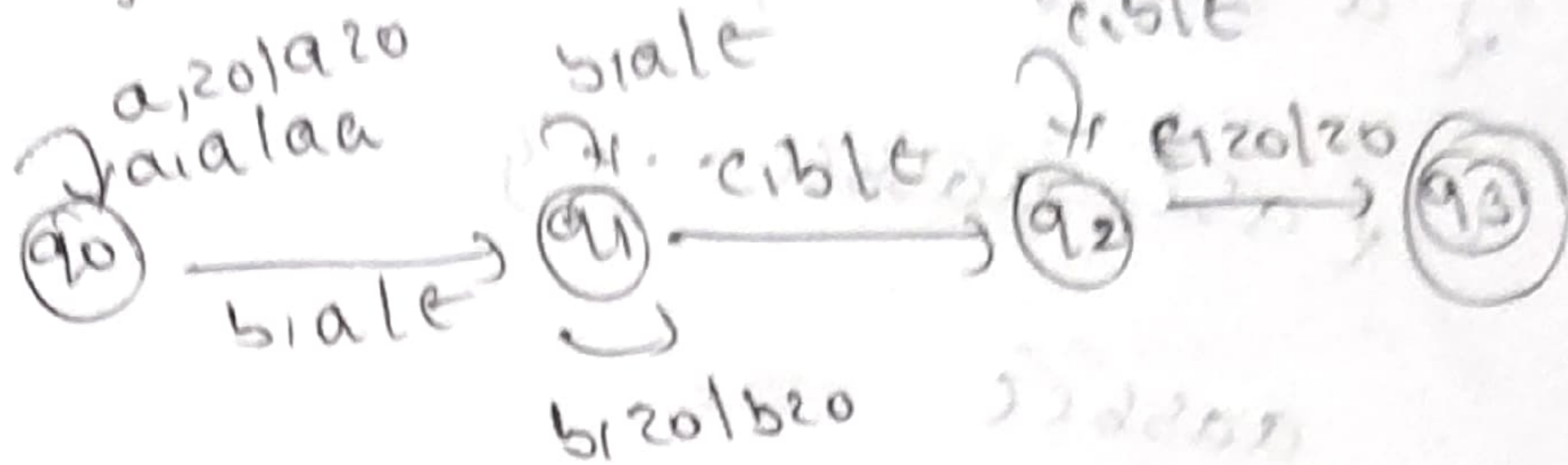


$L = a^n b^{mn} c^{m+1} \mid m \geq 1$

$a^n b^n c^{n+1}$

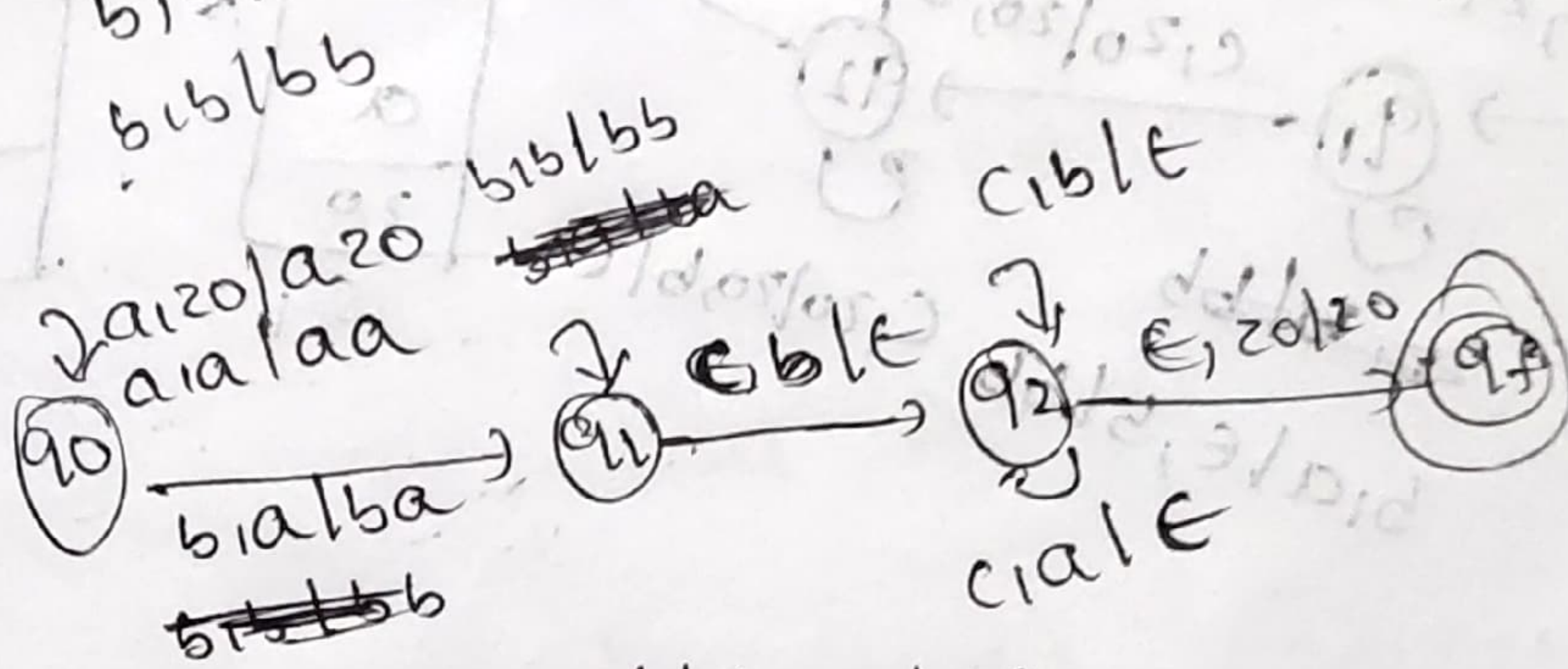
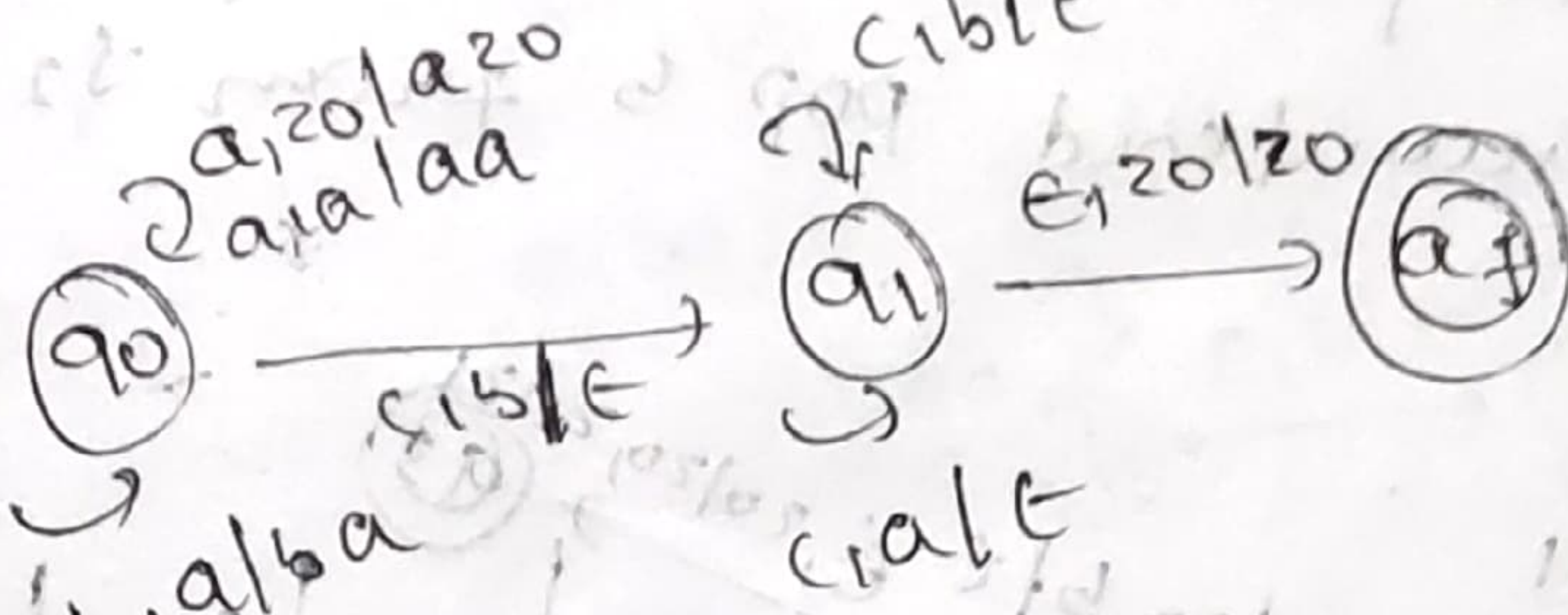
push n a's pop n a's, to show not context free

push m b's, pop m b's, to show not context free

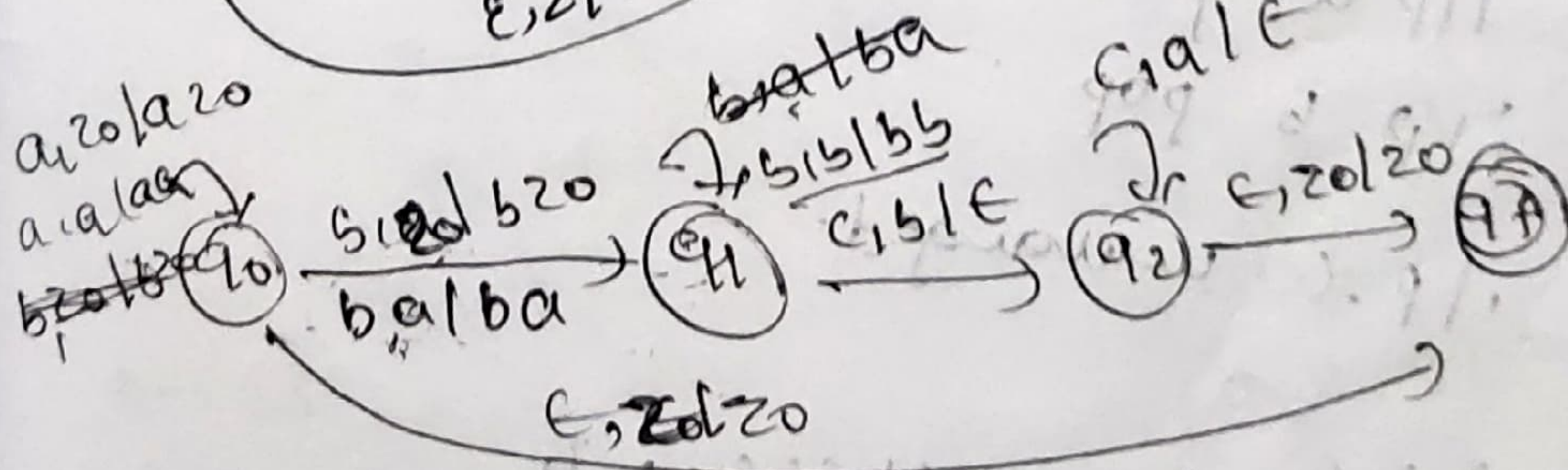
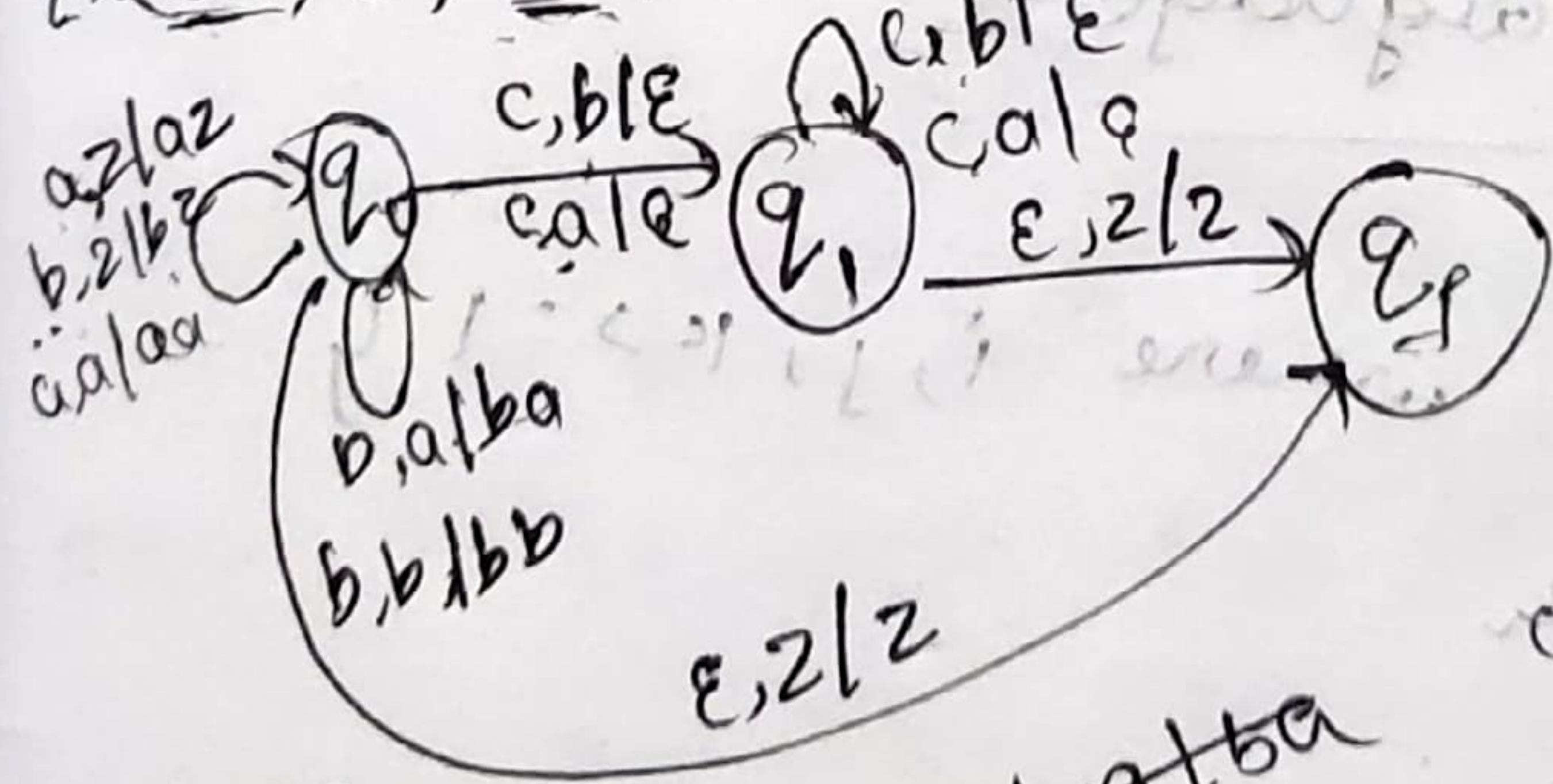


how to show it is not context free? $M = 2$, $n = 1$ $abbbccc$

$L = a^n b^{mn} c^{m+1} \mid m \geq 1$



$L = \{ac, \epsilon, bc, aacc, bbcc, abcc\}$



Two Stack PDA

$$M = (Q, \Sigma, \Gamma, \delta, q_0, z_0, F)$$

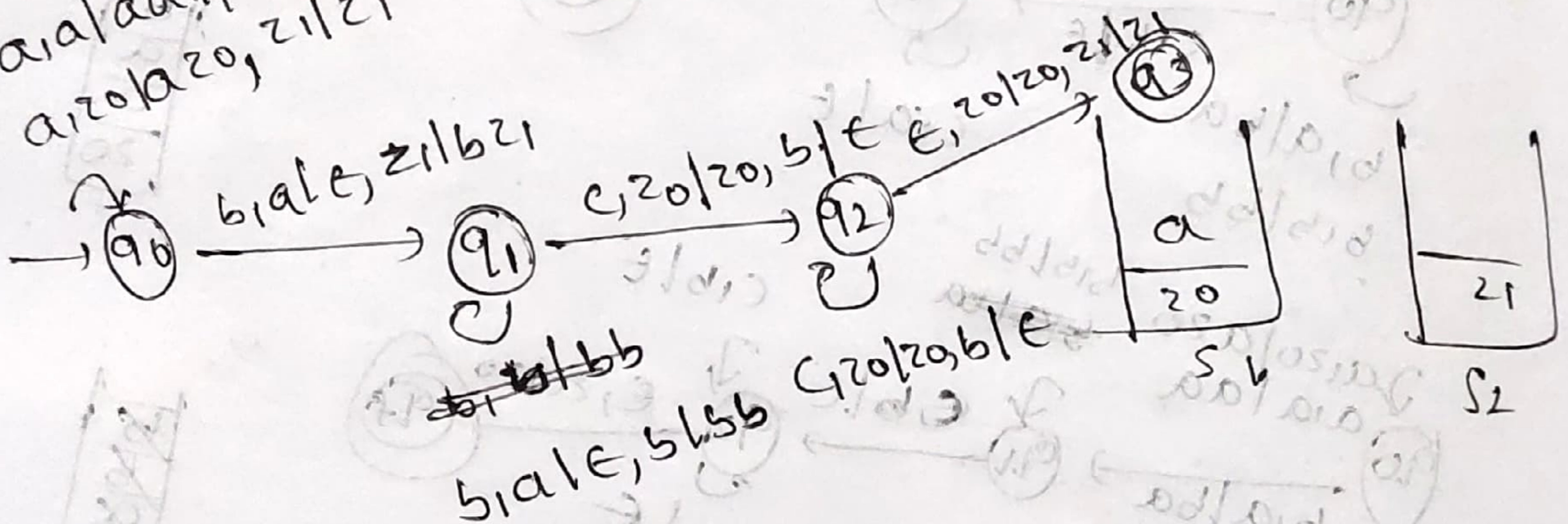
$$L = \{ a^n b^n c^n \mid n \geq 1 \} \rightarrow \text{1 stack not enough}$$

$$L = \{ abc, aabbcc, \dots \}$$

aabbcc

logic:- first put a's into stack 1 and
when b encountered pop a from S1
and push b to S2 and when c
encountered pop b from S2.

a/a a/a z1/z1
a/z0 b/z1



constraint PDA for language

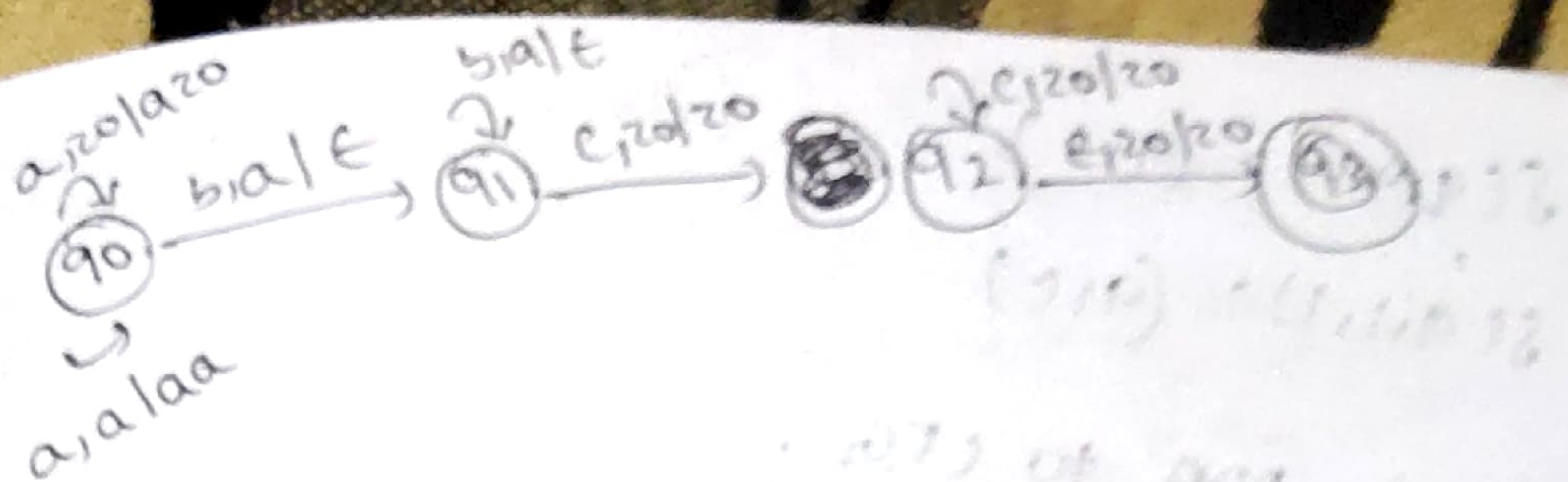
$$L = \{ a^i b^j c^k \mid i = j \text{ where } i, j, k \geq 1 \}$$

logic:- i/p a push

i/p b pop

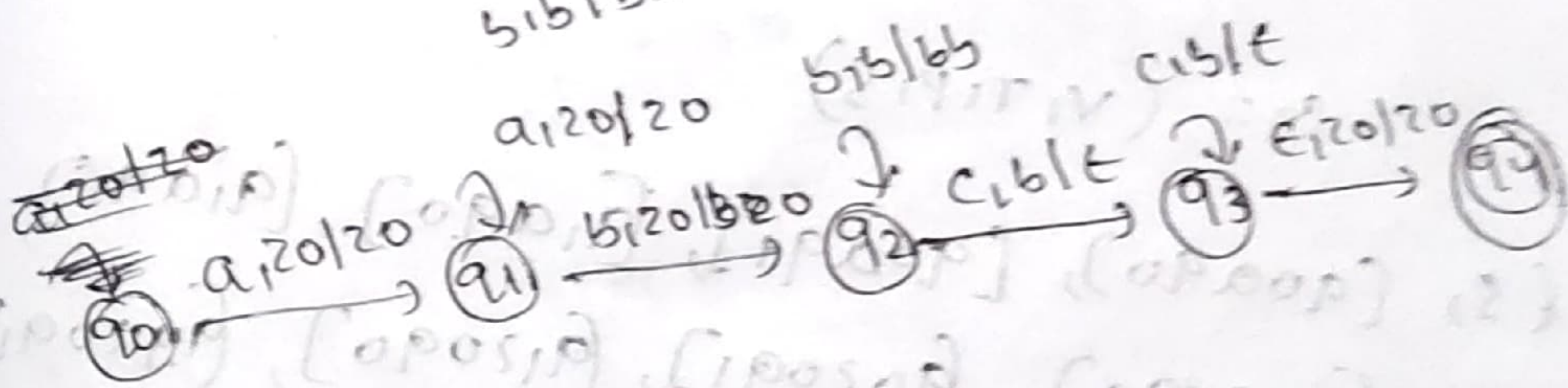
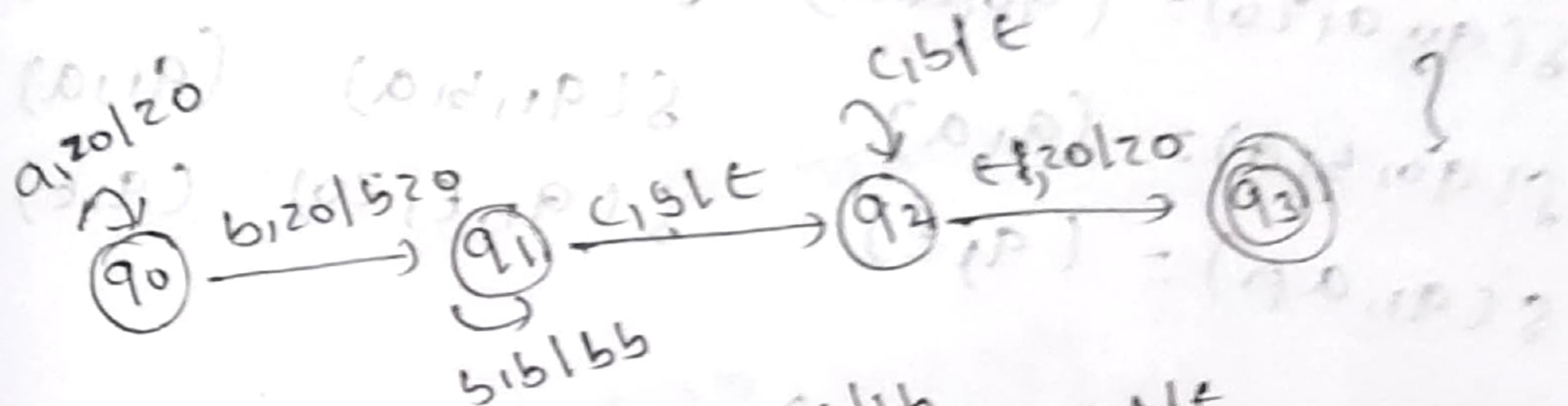
i/p c unchanged.

$$L = \{ abc, aabbcc, \dots \}$$



Q) $L = \{a^i b^j c^k \mid j \leq k, \text{ where } i, j, k \geq 1\}$

(00101) $L = \{abc, abbc, aabc, \dots\}$



conversion of CFG to PDA :-

1) for non terminal, $A \rightarrow \alpha$

$$\delta(q_1, \epsilon, A) = (q_1, \alpha)$$

2) for terminal,

$$\delta(q_1, a, a) = (q_1, \epsilon)$$

Ex: write PDA for CFG

$$S \rightarrow AB$$

$$S \rightarrow AB$$

$$A \rightarrow 0S10$$

$$\delta(q_1, \epsilon, S) = (q_1, AB)$$

$$B \rightarrow 1S1$$

$$A \rightarrow 0S$$

$$\delta(q_1, \epsilon, A) = (q_1, 0S)$$

$$\delta(q_1, \epsilon, B) = (q_1, 1S)$$

$$A \rightarrow 0$$

$$\delta(q_1, \epsilon, A) = (q_1, 0)$$

$$B \rightarrow 1$$

$$\delta(q_1, \epsilon, B) = (q_1, 1)$$

$$\delta(q_0, \epsilon) = (q_1, \epsilon)$$

$$\delta(q_1, 1) = (q_1, \epsilon)$$

convert PDA to CFG !

$$M = (\{q_0, q_1\}, \{a, b\}, \{a, z, \emptyset\}, \delta, q_0, z, \emptyset)$$

$$\delta(q_0, a, z) = (q_0, az)$$

$$\delta(q_0, a, a) = (q_0, aa)$$

$$\delta(q_0, b, a) = (q_1, a)$$

$$\delta(q_1, b, a) = (q_1, a)$$

$$\delta(q_1, \epsilon, z) = (q_1, \epsilon)$$

$$\delta(q_1, a, \epsilon) = (q_1, \epsilon)$$

$$V = \{q_0, q_1, S\}$$

$$V = (S, [q_0 a q_0], [q_0 a q_1], [q_1 a q_0], [q_1 a q_1], [q_0 z a q_0], [q_0 z a q_1], [q_1 z a q_0], [q_1 z a q_1])$$

$$T = \{a, b\}$$

$$S \rightarrow [q_0 z a q_0]$$

$$S \rightarrow [q_0 z a q_1]$$

$$\delta(q_0, a, z) = (q_0, az)$$

$$[q_0 z a q_0] \rightarrow a [q_0 a q_0] [q_0 z a q_0]$$

$$[q_0 z a q_0] \rightarrow a [q_0 a q_1] [q_1 z a q_0]$$

$$[q_0 z a q_1] \rightarrow a [q_0 a q_0] [q_0 z a q_1]$$

$$[q_0 z a q_1] \rightarrow a [q_0 a q_1] [q_1 z a q_1]$$

$$s(q_0, a, a) = (q_0, aa)^{2^2} = (4p \text{ mod})$$

$$[q_0 a q_0] \rightarrow a [q_0 a q_0] [a_0 a a_0]$$

$$[q_0 a q_0] \rightarrow a [q_0 a q_1] [a_1 a q_0]$$

$$[q_0 a q_1] \rightarrow a [q_0 a q_0] [q_0 a q_1]$$

$$[q_0 a q_1] \rightarrow a [q_0 a q_1] [a_1 a q_1]$$

$$s(q_0, a, a) = (q_1, a)$$

$$[q_0 a q_0] \rightarrow b [q_1 a q_0]$$

$$[q_0 a q_1] \rightarrow b [q_1 a q_1]$$

$$s(q_1, b, a) = (q_1, a)$$

$$[q_1 a q_0] \rightarrow b [q_1 a q_0]$$

$$[q_1 a q_1] \rightarrow b [q_1 a q_1]$$

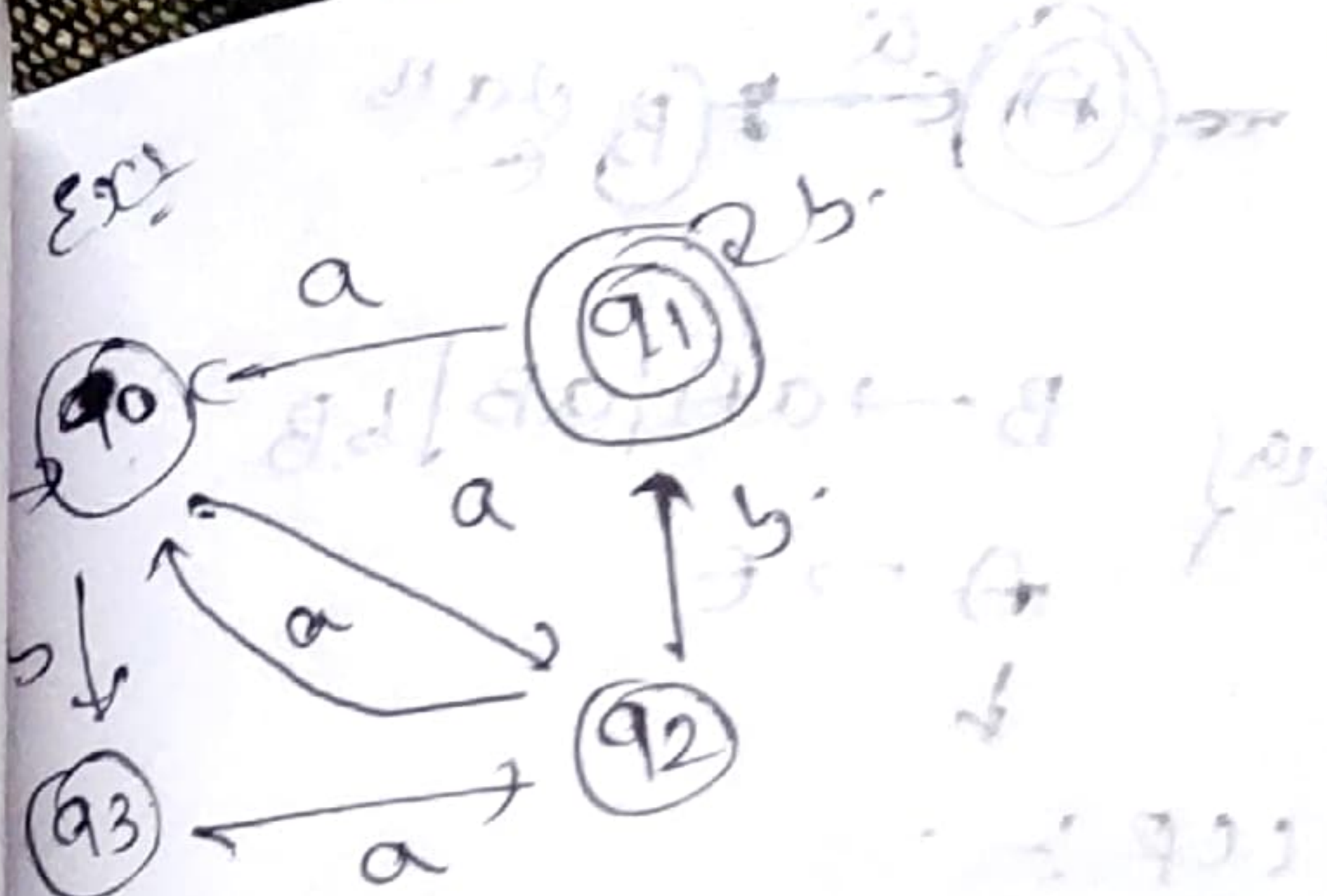
$$s(q_1, a, a) = (q_1, \epsilon) \quad 2^0 = 1 \text{ if } \epsilon \text{ or } R_H \text{ only terminal}$$

$$[q_1 a q_1] \rightarrow \overline{[q_1 a q_1]} a$$

$$s(q_1, \epsilon, \epsilon) = (q_1, \epsilon)$$

$$[q_1 \epsilon q_1] \rightarrow \overline{[q_1 \epsilon q_1]}$$

ϵ



$G = (V, T, P, S)$
 $V = \{q_0, q_1, q_2, q_3\}$
 $T = \{a, b\}$
 $S = q_0$

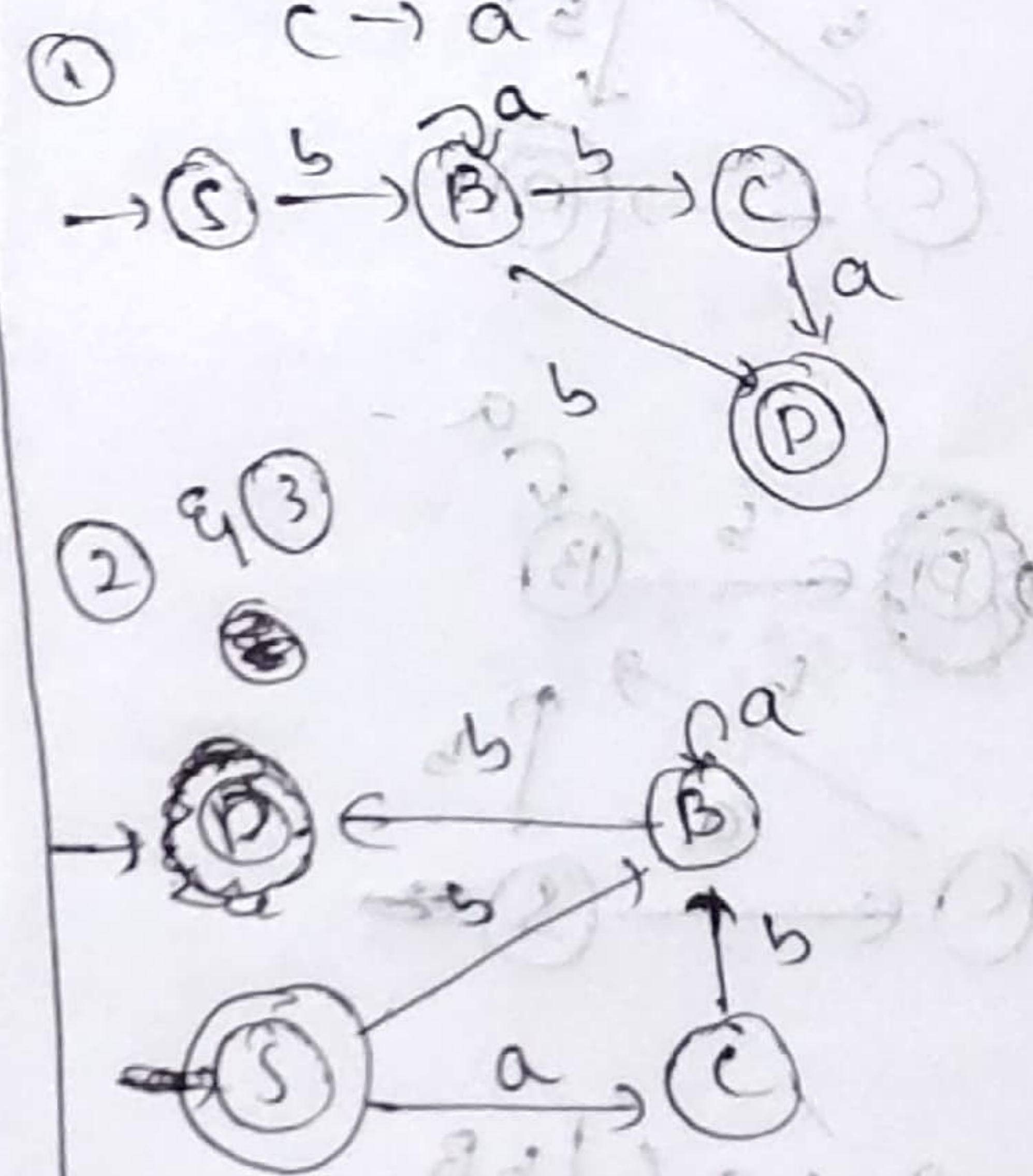
prod rules,

- $q_0 \rightarrow a q_1$
- $q_0 \rightarrow b q_3$
- $q_1 \rightarrow a q_0$
- $q_1 \rightarrow b q_2$
- $q_2 \rightarrow a q_3$
- $q_2 \rightarrow b q_1$
- $q_3 \rightarrow a q_2$
- $q_3 \rightarrow b q_0$

converting RL to LL

- 1) const. FA for given RL
- 2) Interchange initial & final state
- 3) Reverse the direction on all edges
- 4) const. regular grammar from FA

Ex: $S \rightarrow bB$
 $B \rightarrow bC | aB | b$
 $C \rightarrow a$

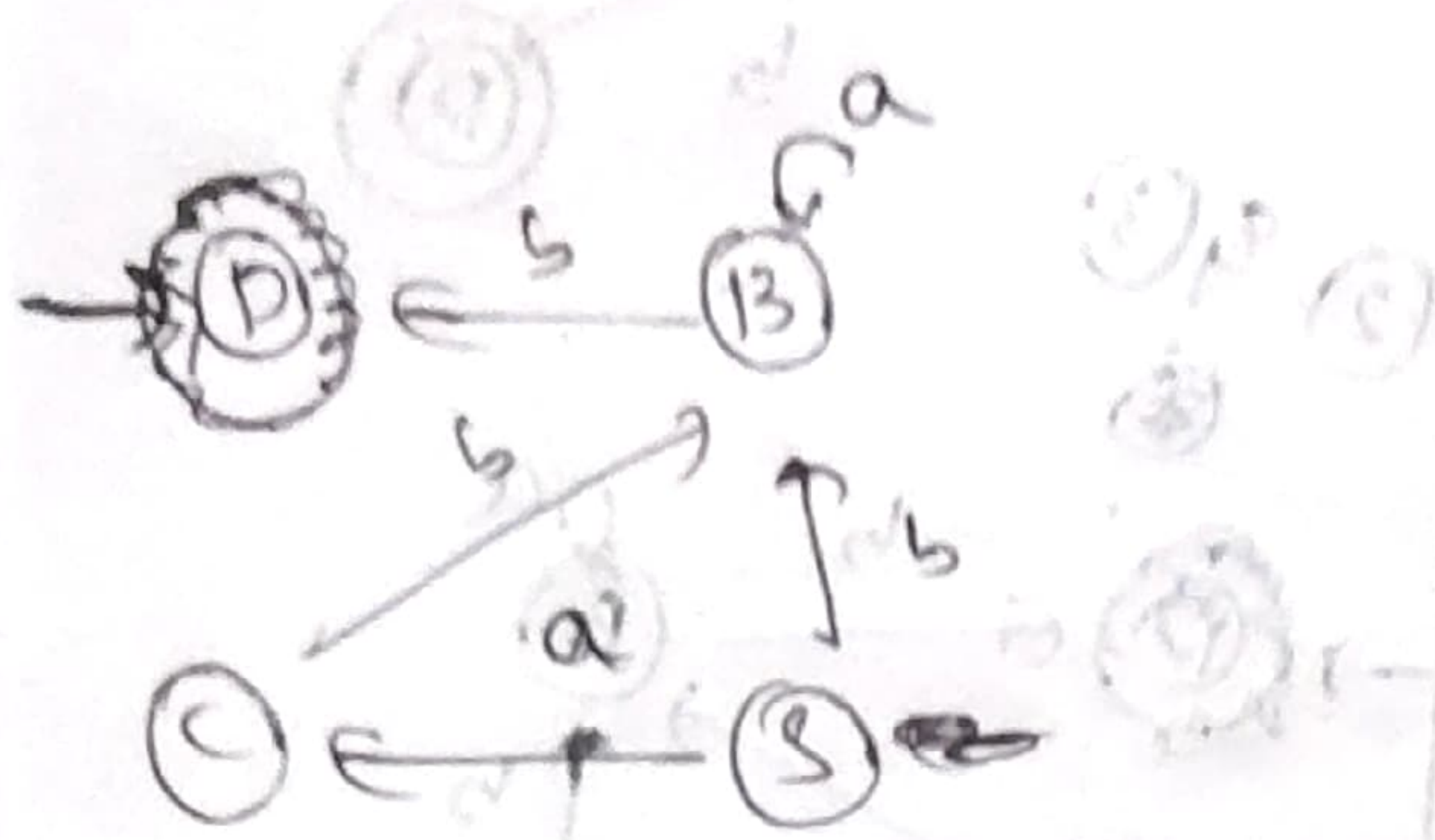
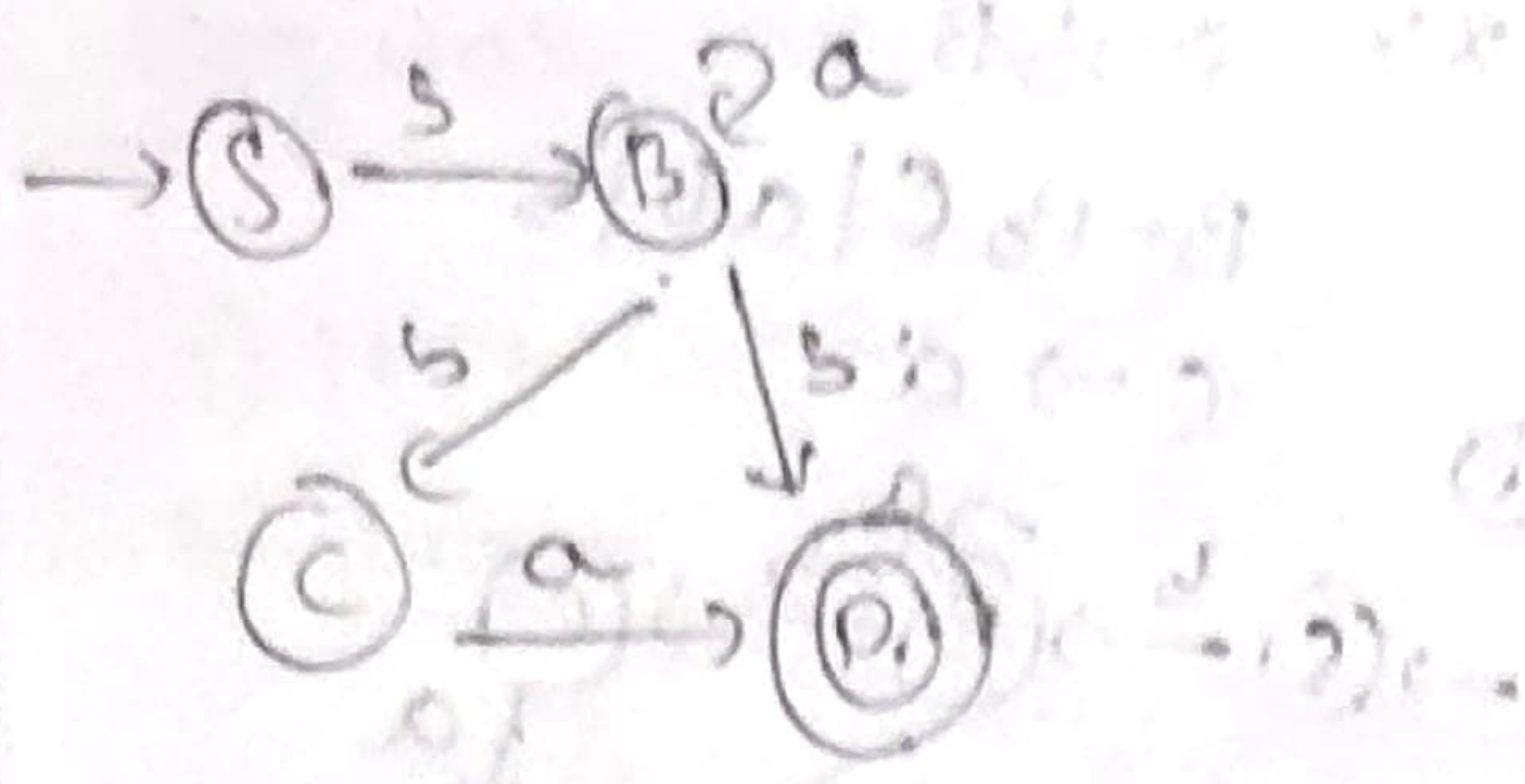


$S \rightarrow aC | bB$
 $B \rightarrow aB | bD | b$
 $C \rightarrow bB$

from FA we can obtain only RLG, after we const. RLG we interchange pos of terminals & non-terminals.

$S \rightarrow Ca | Bb$
 $B \rightarrow Ba | Db | b$
 $C \rightarrow Bb$

Ex: $S \rightarrow bB$
 $B \rightarrow bC$
 $B \rightarrow aB$
 $C \rightarrow a$
 $B \rightarrow b$



$$S \rightarrow aC | bB$$

$$C \rightarrow bB$$

$$B \rightarrow aB | bD | b$$

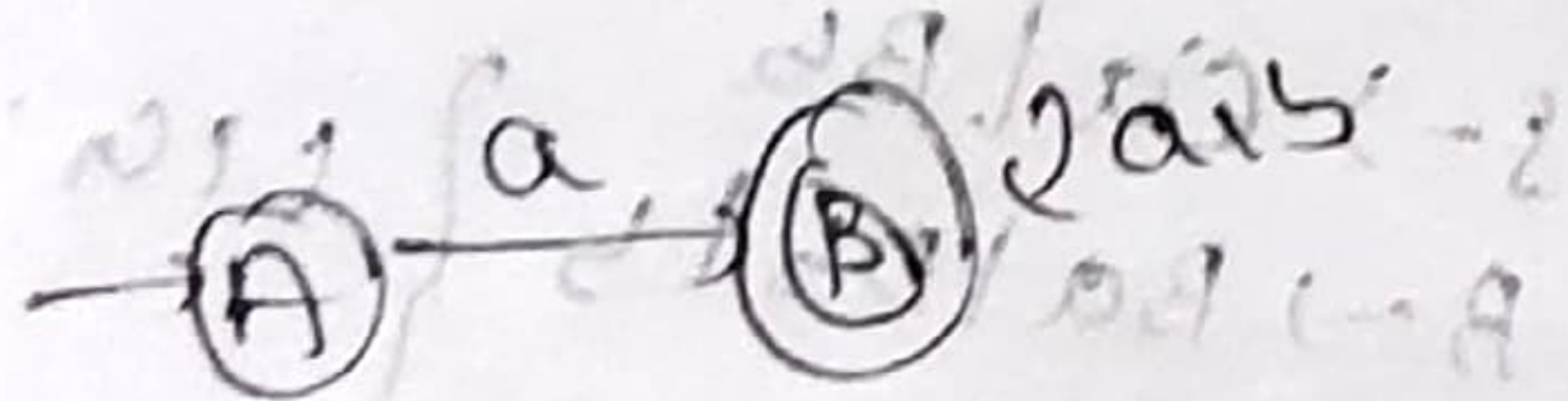
$$S \rightarrow aC | bB$$

$$C \rightarrow bB$$

$$B \rightarrow aB | bD | b$$

convert FA to RLG or

FA to LLG



RLG

$$A \rightarrow aB$$

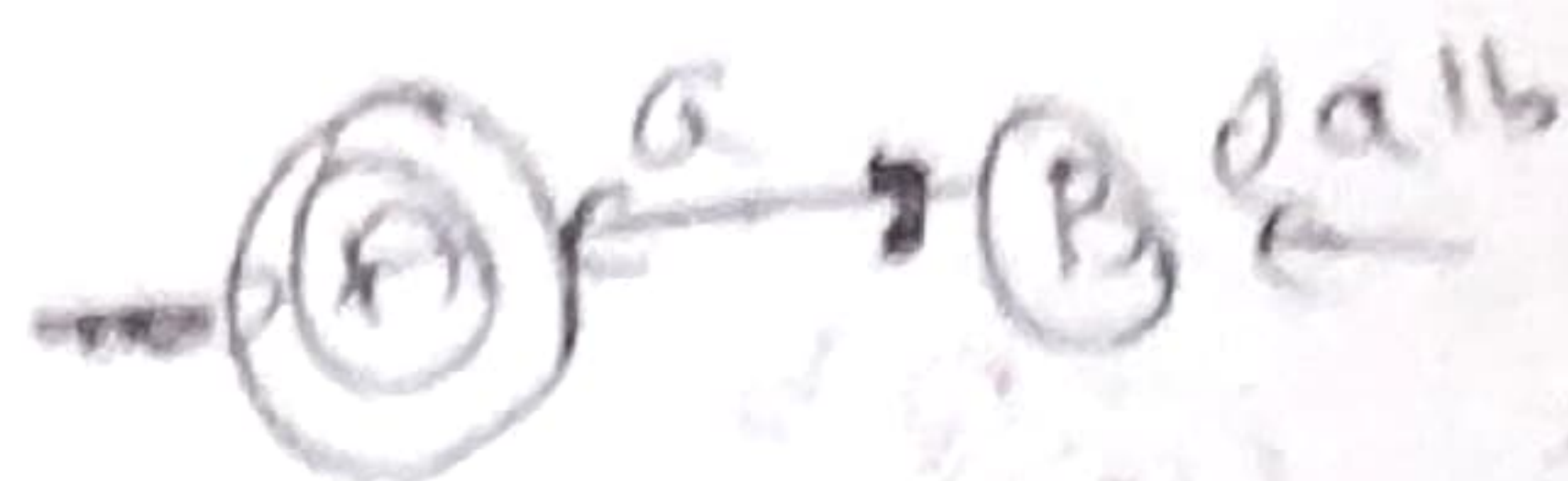
$$B \rightarrow aB | bB$$

$$B \rightarrow \epsilon$$

1) Reverse FA

2) convert RLG

3) Reverse RLG



$$\begin{aligned} B &\rightarrow aA | aB | bB \\ A &\rightarrow \epsilon \end{aligned}$$

LLG:

$$B \rightarrow aA | aB | bB$$

$$A \rightarrow \epsilon$$

convert FA for RLG

or

convert RLG to FA

procedure: let $G = (V, T, \delta, q_0, F)$

be RLG $m = (Q, \Sigma, \delta, q_0, F)$

Q corresponds to V

ϵ corresponds to T

q_0 corresponds to S

F corresponds to Σ

1) If production is in form $A_i \rightarrow aA_j$ then transition function $\delta(A_i, a) = A_j$

2) If a prod is in form $A_i \rightarrow a$ then introduce final state $S(A_i, a) = A_j$ where A_j final state.

$$\text{Ex: } S \rightarrow aA | aA$$

$$A \rightarrow aA | aA | bB | bB$$

$$B \rightarrow aB | bB | \epsilon$$

$$G = (V, \tau, P(S))$$

$$V = \{S, A, B\}$$

$$\tau = \{0, 1, +, -\}$$

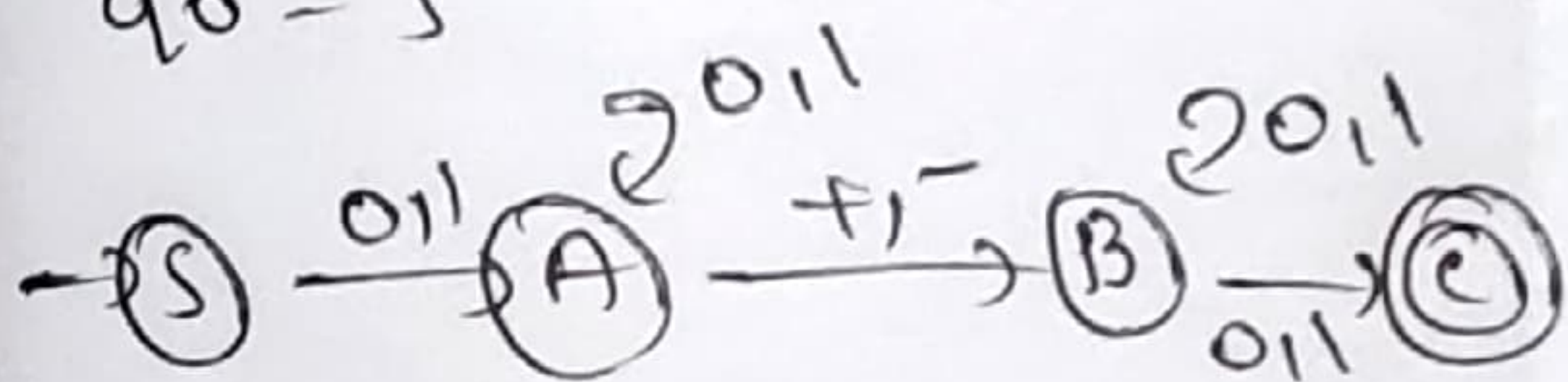
$$S \rightarrow \{S\}$$

$$M = (Q, E, \delta, q_0, f)$$

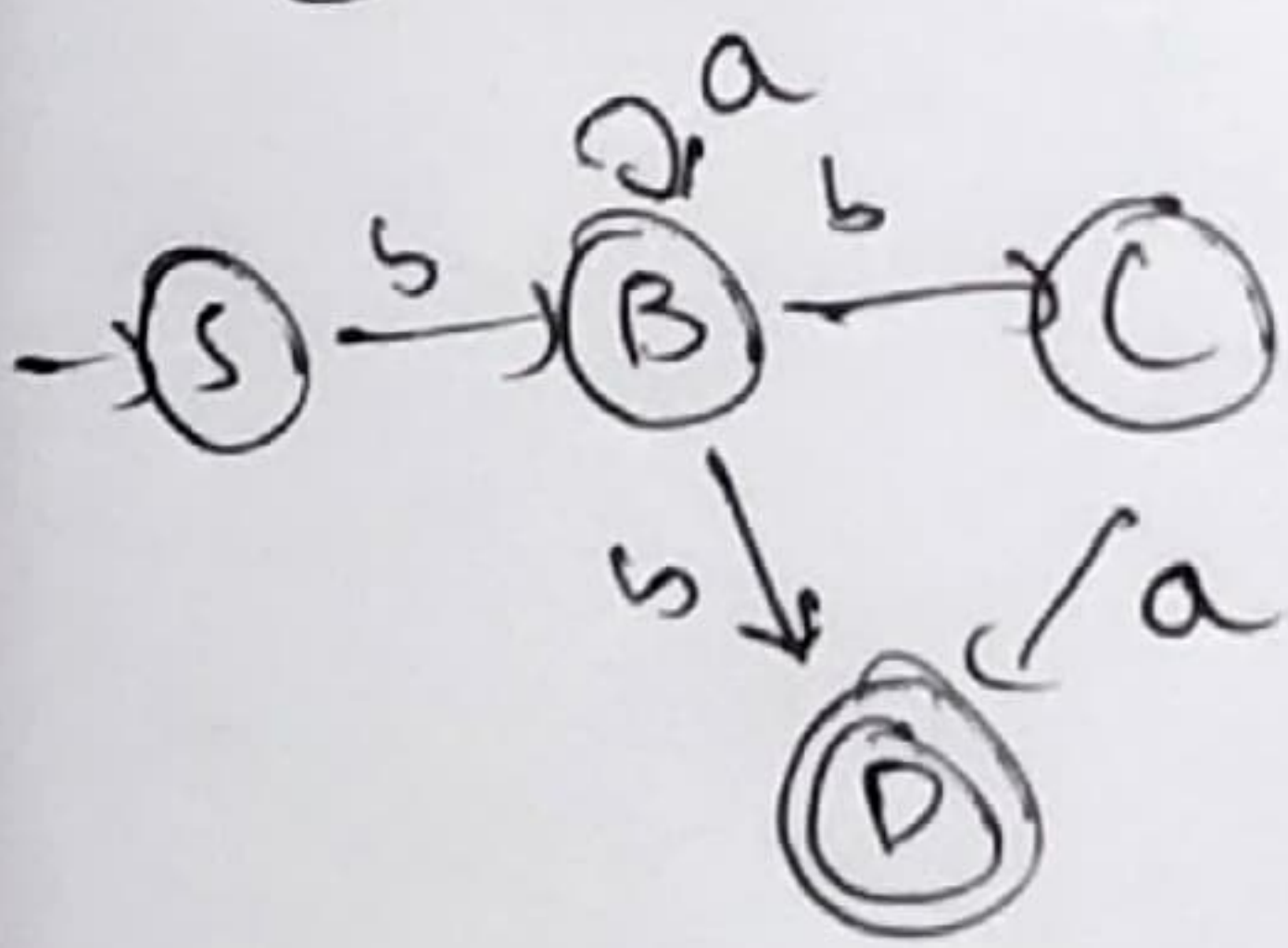
$$Q = \{S, A, B\}$$

$$E = \{0, 1, +, -\}$$

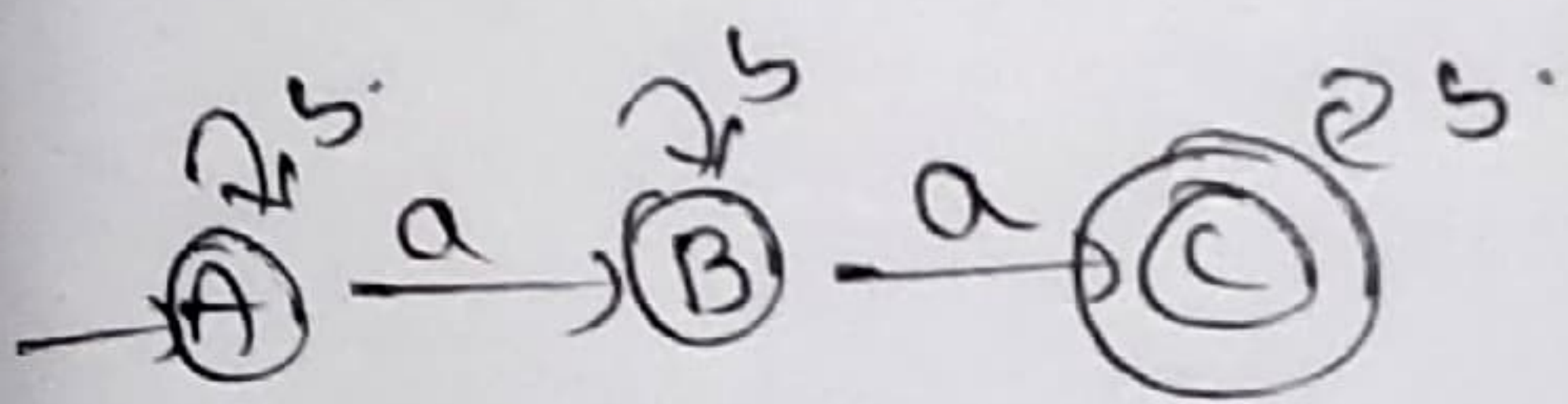
$$q_0 = S$$



2) $S \rightarrow bB$
 $B \rightarrow bC | aB | b$
 $C \rightarrow a$



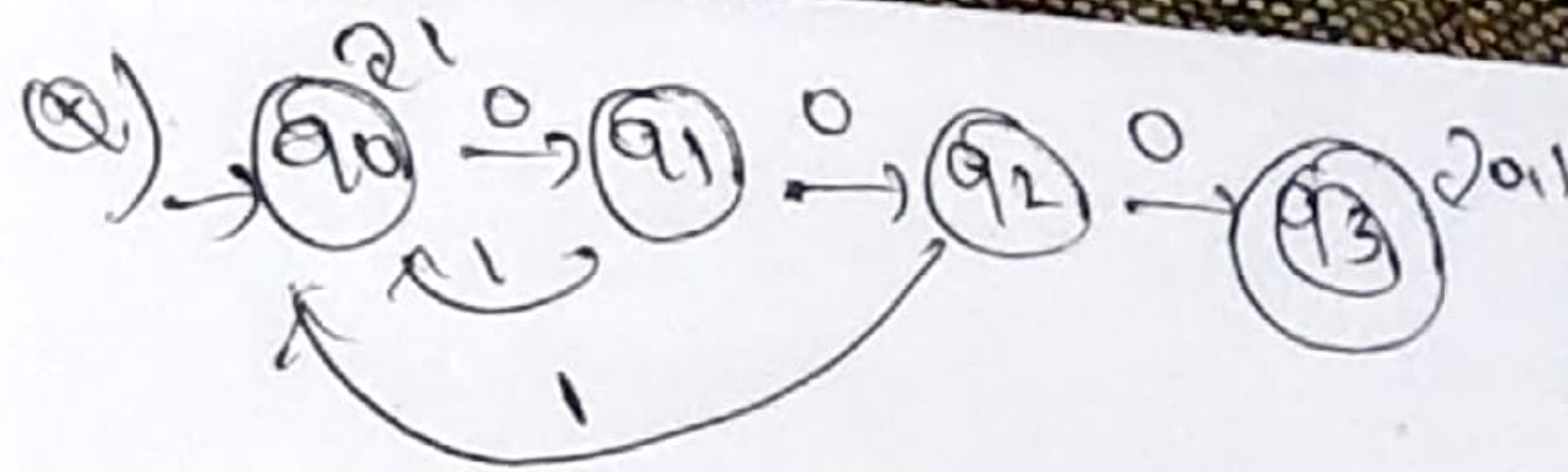
FA to RG questions :



$$A \rightarrow bA | aB$$

$$B \rightarrow bB | aC | a$$

$$C \rightarrow bC | b$$



$$q_0 \rightarrow 1q_0 | 0q_1$$

$$q_1 \rightarrow 0q_2 | 1q_0$$

$$q_2 \rightarrow 0q_3 | 1q_0 | 0$$

$$q_3 \rightarrow 0q_3 | 1q_3 | 0 | 1$$